

**GREEN
CLIMATE
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Meeting of the Board
27 February – 1 March 2018
Songdo, Incheon, Republic of Korea
Provisional agenda item 15

GCF/B.19/22/Add.17

6 February 2018

Consideration of funding proposals – Addendum XVII

Funding proposal package for FP075

Summary

This addendum contains the following three parts:

- a) A funding proposal summary titled “Institutional Development of the State Agency for Hydrometeorology of Tajikistan”;
- b) No-objection letters issued by the national designated authority(ies) or focal point(s); and
- c) Environmental and social report(s) disclosure;



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Funding Proposal

Version 1.1

The Green Climate Fund (GCF) is seeking high-quality funding proposals.

Accredited entities are expected to develop their funding proposals, in close consultation with the relevant national designated authority, with due consideration of the GCF's Investment Framework and Results Management Framework. The funding proposals should demonstrate how the proposed projects or programmes will perform against the investment criteria and achieve part or all of the strategic impact results.

Project/Programme Title: Institutional Development of the State Agency for Hydrometeorology of Tajikistan

Country/Region: Tajikistan

Accredited Entity: Asian Development Bank

Date of Submission: 16 October 2017

Contents

Section A	PROJECT / PROGRAMME SUMMARY
Section B	FINANCING / COST INFORMATION
Section C	DETAILED PROJECT / PROGRAMME DESCRIPTION
Section D	RATIONALE FOR GCF INVOLVEMENT
Section E	EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA
Section F	APPRAISAL SUMMARY
Section G	RISK ASSESSMENT AND MANAGEMENT
Section H	RESULTS MONITORING AND REPORTING
Section I	ANNEXES

Note to accredited entities on the use of the funding proposal template

- Sections **A, B, D, E** and **H** of the funding proposal require detailed inputs from the accredited entity. For all other sections, including the Appraisal Summary in section F, accredited entities have discretion in how they wish to present the information. Accredited entities can either directly incorporate information into this proposal, or provide summary information in the proposal with cross-reference to other project documents such as project appraisal document.
- The total number of pages for the funding proposal (excluding annexes) is expected not to exceed 50.

Please submit the completed form to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

"[FP]-[Agency Short Name]-[Date]-[Serial Number]"

A.1. Brief Project / Programme Information		
A.1.1. Project / programme title		Institutional Development of the State Agency for Hydrometeorology of Tajikistan
A.1.2. Project or programme		Project
A.1.3. Country (ies) / region		Tajikistan
A.1.4. National designated authority (ies)		Mr. K. Ibodzoda, Chairman, Committee on Environmental Protection
A.1.5. Accredited entity		Asian Development Bank
A.1.5.a. Access modality		☐ Direct ☒ International
A.1.6. Executing entity / beneficiary		Executing Entity: Ministry of Finance ¹
A.1.7. Project size category (Total investment, million USD)		☒ Micro (≤ 10) ☐ Small ($10 < x \leq 50$) ☐ Medium ($50 < x \leq 250$) ☐ Large (> 250)
A.1.8. Mitigation / adaptation focus		☐ Mitigation ☒ Adaptation ☐ Cross-cutting
A.1.9. Date of submission		16 October 2017
A.1.10. Project contact details	Contact person, position	Nathan Rive, Climate Change Specialist
	Organization	Asian Development Bank (ADB)
	Email address	nrive@adb.org
	Telephone number	+63-2-632-4444
	Mailing address	6 ADB Ave., Mandaluyong City 1550, Philippines

A.1.11. Results areas <i>(mark all that apply)</i>	
<u>Reduced emissions from:</u> ☐ Energy access and power generation (E.g. on-grid, micro-grid or off-grid solar, wind, geothermal, etc.) ☐ Low emission transport (E.g. high-speed rail, rapid bus system, etc.) ☐ Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.) ☐ Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural irrigation, water treatment and management, etc.)	
<u>Increased resilience of:</u> ☒ Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)	

¹ ADB will conclude a financing agreement with the Republic of Tajikistan represented by the Ministry of Finance on the management and use of proceeds. The Ministry of Finance will cause the project to be implemented through the State Agency for Hydrometeorology (Hydromet).

- ☐ Health and well-being, and food and water security
(E.g. climate-resilient crops, efficient irrigation systems, etc.)
- ☐ Infrastructure and built environment
(E.g. sea walls, resilient road networks, etc.)
- ☐ Ecosystem and ecosystem services
(E.g. ecosystem conservation and management, ecotourism, etc.)

A.2. Project / Programme Executive Summary (max 300 words)

Please provide a brief description of the proposed project/programme, including the objectives and primary measurable benefits (see [investment criteria in section E](#)). The detailed description can be elaborated in [section C](#).

Tajikistan already faces recurring climate-related extreme weather events, particularly flood and landslides. Climate change is expected to exacerbate these impacts^{2,3}. Districts in the Pyanj River Basin, in particular, are among poorest and most vulnerable to climate change⁴.

However, low capacity in the country the forecasting of weather, hydrometeorological, and climatic conditions remain a critical weakness in the country. Disaster risk mitigation and response in vulnerable communities, as well as broader climate change adaptation efforts, are thus hampered. The responsible forecasting entity, the State Agency for Hydrometeorology (Hydromet) is crippled by many of the common problems faced by post-Soviet institutions including limited budget, decaying infrastructure, and poor staff retention. Its technical capacity is relatively low, and government and civil society stakeholders are generally unsatisfied with the forecasting services offered by Hydromet. Without interventions addressing its underlying weaknesses, Hydromet capacity and outputs will likely continue to languish.

The subject proposed project will support Hydromet's development to a sustainable and well-resourced institution that produces timely and accurate forecasting of extreme weather events particularly in the PRB. In doing so, the project will address key institutional weaknesses. The project will (i) modernize Hydromet's campus and associated facilities [Component A], (ii) support legal and organizational transformation and capacity building of Hydromet into a government entity with flexibility to set staff salaries and retain additional entrepreneurial revenue [Component B], (iii) undertake capacity building for improve forecasting and warning of extreme weather events [Component C], and (iv) support development and implementation of a viable business model through marketing of fee-based services [Component D]. The project's primary measurable benefit (indicators) are (i) Hydromet develops and disseminates timely and accurate forecasts of flood events in the PRB and (ii) Hydromet develops a weather service product marketed to customers in the PRB.

Impact: The project will have direct benefits to individuals, community organizations, and government agencies in the Pyanj River Basin, and broader indirect benefits to the wider country from a strengthened Hydromet agency.

Paradigm shift potential: The project will facilitate a significant shift in the funding and provision of information and forecasting services in Tajikistan. The model is replicable in other hydrometeorology agencies around Central Asia. The project will help enable the scaling up of Hydromet's products and services into for example agrometeorology.

Sustainable development potential: The project will provide direct and indirect economic and social benefits to the PRB, wider country, and Hydromet. This includes (i) more attractive and well-paid jobs in Hydromet, (ii) improved disaster preparedness and response, and (iii) increased productivity in key economic sectors.

² Murnane, et al., 2016. *Future scenarios for earthquake and flood risk in Eastern Europe and Central Asia*. Earth's Future. 5(7): 693-714.

³ Westphal, 2008. *Summary of Climate Science in the Europe and Central Asia Region: Historical Trends and Future Projections*. World Bank. Washington DC.

⁴ ADB, 2011. *Climate Resiliency for Natural Resources Investments*. TA Consultant Final Report. Manila.

Needs of recipient: Tajikistan is highly vulnerable to climate change, and communities in the PRB are among the most exposed, vulnerable, and poor in the country.

Country ownership: The Tajikistan Nationally Determined Contribution (NDC) to the UNFCCC Paris Agreement prioritizes improvement and modernization of hydrometeorological services as a key climate change adaptation. The Government of Tajikistan has requested ADB's support to transform and modernize Hydromet.

Efficiency and effectiveness: GCF co-financing ratio is 50%. Economic internal rate of return (EIRR) of the project is estimated at 29% and robust to key cost and benefit sensitivities.

A.3. Project/Programme Milestone

Expected approval from accredited entity's Board (if applicable)	31/03/2018
Expected financial close (if applicable)	Q4 2023
Estimated implementation start and end date	Start: Q3 2018 End: Q2 2023
Project/programme lifespan	Implementation period: 5 years Project benefit stream: 30 years

B.1. Description of Financial Elements of the Project / Programme

The project's financing plan by financier is provided in Table 1 below.

Table 1: Project financing by financier (USD million)

Source	Instrument	Overall Project (USD million)	Share (%)
ADB	Grant	5.0	45%
GCF	Grant	5.0	45%
Government	Exemptions ⁵	1.1	10%
Total		11.1	100%

Source: Asian Development Bank.

ADB will contribute \$5.0 million grant financing from the Asian Development Fund under the disaster risk reduction funding mechanism⁶ and an equal grant is requested from GCF. Detailed budget by expenditure and activities are provided in Table 2 and Table 3.

Table 2 : Project budget by expenditure item (USD)

	ADB (ADF DRR)	GCF	Total ^a	% Total Base Cost
A. Investment Cost				
1 Civil Works				
a. Main office and ancillary		1,495,750	1,495,750	17%
b. Mixed-use buildings	3,734,831		3,734,831	41%
2 Equipment	162,500	949,300	1,111,800	12%
3 Survey, Study & Design	112,500	53,500	166,000	2%
4 Training & Workshops	26,875	123,125	150,000	2%
5 Consulting Services	324,078	1,198,519	1,522,597	17%
Subtotal	4,360,784	3,820,193	8,180,978	91%
B. Recurrent Cost				
1 PMO Salaries	107,214	390,786	498,000	6%
2 O&M Equipment	3,125	171,856	174,981	2%
3 Office / Vehicle O&M	33,381	121,669	155,050	2%
Subtotal	143,720	684,311	828,031	9%
Total Base Cost	4,504,505	4,504,505	9,009,009	100%
C. Contingencies				
1 Physical	225,225	225,225	450,450	5%
2 Price	270,270	270,270	540,541	6%
Subtotal	495,495	495,495	990,991	11%
Total	5,000,000	5,000,000	10,000,000	111%

^a Plus government in-kind contribution of taxes and duties forgone and social costs estimated at \$1.141 million.

ADB (ADF DRR) = Asian Development Bank Asian Development Fund Disaster Risk Reduction mechanism; GCF = Green Climate Fund

Source: Asian Development Bank

⁵ Estimated taxes and duties foregone and social costs.

⁶ ADB, 2016. *Concessional Assistance Policy 2016*. Manila.

**Table 3 : Project budget by component including contingencies
(USD million)**

	ADB (ADF DRR)	GCF	Total ^a	GCF Share (%)
Component A - Campus modernization	4,446,541	2,348,512	6,795,053	35%
1.1 Design and supervision services engaged	300,878	365,730	666,608	55%
1.2 Core operational systems modernized	0	1,982,782	1,982,782	100%
1.3 Mixed-use buildings constructed	4,145,663	0	4,145,663	0%
Component B - Legal and organizational transformation	179,541	511,764	691,305	74%
1.3 Legal reforms implemented	39,682	110,989	150,672	74%
1.4 Institutional framework and quality management system enhanced	75,812	216,692	292,503	74%
1.5 Capacity building and knowledge-sharing program implemented	64,047	184,083	248,130	74%
Component C - Forecasting and warning of extreme weather	373,918	1,094,894	1,468,812	75%
1.6 Community awareness and preparedness enhanced	68,892	198,619	267,511	74%
1.7 Flood forecasting system developed and capacity building undertaken	305,026	896,275	1,201,301	75%
Component D - Marketing of fee-based services	0	1,044,830	1,044,830	100%
3.1 Community awareness and weather product accessibility enhanced	0	289,448	289,448	100%
3.2 Subscription platform and new weather products developed	0	755,382	755,382	100%
Total	5,000,000	5,000,000	10,000,000	50%

^a Plus government in-kind contribution of taxes and duties forgone and social costs estimated at \$1.141 million.

ADB (ADF DRR) = Asian Development Bank Asian Development Fund Disaster Risk Reduction mechanism; GCF = Green Climate Fund

Source: Asian Development Bank

Grant finance is requested as Hydromet is not a profit-making enterprise and is expected to continue to rely on core government funding for the foreseeable future. In addition, Tajikistan is a country with high level of debt distress⁷.

B.2. Project Financing Information

	Financial Instrument	Amount	Currency	Tenor	Pricing
(a) Total project financing	(a) = (b) + (c)	10	million USD (\$)		
(b) GCF financing to recipient	(i) Senior Loans		Options	() years	() %
	(ii) Subordinated Loans		Options	() years	() %
	(iii) Equity		Options		() % IRR
	(iv) Guarantees		Options		
	(v) Reimbursable grants *		Options		
	(vi) Grants *	5.0	million USD (\$)		
* Please provide economic and financial justification in section F.1 for the concessionality that GCF is expected to provide, particularly in the case of grants. Please specify difference in tenor and price between GCF financing and that of accredited entities. Please note that the level of c.2.2.concessionality should correspond to the level of the project/programme's expected performance against the investment criteria indicated in section E .					

⁷ <https://www.imf.org/external/Pubs/ft/dsa/DSAlist.pdf>

	Total requested (i+ii+iii+iv+v+vi)		5.0		million USD (\$)			
(c) Co-financing to recipient	Financial Instrument	Amount	Currency	Name of Institution	Tenor	Pricing	Seniority	
	<u>Grant</u>	5.0	<u>million USD (\$)</u>	ADB	() years	() %	<u>Options</u>	
	<u>Options</u>		<u>Options</u>		() years	() %	<u>Options</u>	
	<u>Options</u>		<u>Options</u>		() years	() % IRR	<u>Options</u>	
	<u>Options</u>		<u>Options</u>				<u>Options</u>	
Lead financing institution: ADB								
Plus government in-kind contribution of taxes and duties forgone and social costs estimated at \$1.141 million.								
* Please provide a confirmation letter or a letter of commitment in section I issued by the co-financing institution.								
(d) Financial terms between GCF and AE (if applicable)	<i>In cases where the accredited entity (AE) deploys the GCF financing directly to the recipient, (i.e. the GCF financing passes directly from the GCF to the recipient through the AE) or if the AE is the recipient itself, in the proposed financial instrument and terms as described in part (b), this subsection can be skipped.</i> <i>If there is a financial arrangement between the GCF and the AE, which entails a financial instrument and/or financial terms separate from the ones described in part (b), please fill out the table below to specify the proposed instrument and terms between the GCF and the AE.</i>							
	Financial instrument	Amount	Currency	Tenor	Pricing			
			<u>Options</u>	() years	() %			
	<i>Please provide a justification for the difference in the financial instrument and/or terms between what is provided by the AE to the recipient and what is requested from the GCF to the AE.</i>							
B.3. Financial Markets Overview (if applicable)								
Not applicable.								

Please fill out applicable sub-sections and provide additional information if necessary, as these requirements may vary depending on the nature of the project / programme.

C.1. Strategic Context

Please describe relevant national, sub-national, regional, global, political, and/or economic factors that help to contextualize the proposal, including existing national and sector policies and strategies.

Tajikistan is one of the wellsprings of Central Asia; the Syr and Amu Rivers, originating in Tajikistan and Kyrgyz Republic and flowing into Turkmenistan, Uzbekistan, and Kazakhstan, comprise 90% of the water resources in the region. Roughly 70% of the population lives in rural areas and rely on agriculture as a primary source of income. Their livelihoods are thus directly dependent on the country's water resources and exposed to the prevailing climatic conditions and weather events. It is also Central Asia's poorest country and among the region's most vulnerable countries to climate change, as ranked by the ND-GAIN vulnerability index.⁸

Tajikistan is a country highly prone to climate-related extreme weather events, notably flooding, which has caused roughly 80% of disaster mortalities in the country in the period 1990-2016.⁹ Climate change is expected to exacerbate these adverse events and their impacts. Higher temperatures and changes in precipitation patterns are expected to cause earlier and faster snowmelt and recession of glaciers and a decline in overall water availability. Water stress conditions are likely to become more common, and flooding and landslides are likely to become more frequent and damaging¹⁰.

Climate vulnerability is particularly acute in the districts along the Pyanj River Basin (PRB), the primary tributary to the Amu River in the south of the country, which are among the country's poorest and comprise a wide range of geographical and climatic conditions. The PRB is the largest river basin in the country (114,500 km²) spans 18 administrative districts across two administrative regions (Khatlon and Gorno Badakhshan provinces), with a population of approximately 1.3 million.¹¹ The mountain massifs that define the PRB feature glaciers and permanent snow fields at elevations above 3,500 m, and have high density drainage networks of steep alpine streams delivering runoff and sediment rapidly to the alluvial fan. Water resources management in the PRB is critical to the country's economy and development. Khatlon is the province with the largest population (2.7 million) and agriculture production (e.g. 774,000 tons cereal production) in the country, it includes the most food insecure zone among country's irrigated area.¹² PRB is the country's poorest river basin (55% poverty rate).¹³ It is estimated that 14% of the PRB is at high to extremely high risk of floods, and this is expected to increase to 20% by 2070.¹⁴ In Khatlon region alone between 1998-2016, there occurred annually roughly 30 natural disaster events with 4 lives lost and damage exceeding over \$3 million per year.¹⁵ Annual river flows are likely to increase in the glacial sub-basins between 2015 and 2065 or 2075, due to increased melt caused by higher air temperatures. The expected change in monthly flow rates due to increased rainfall and decreased snowfall may entail an increase in the magnitude and frequency of extreme flood events. A gradual shift in the river flow seasonal distribution and increase of irrigation water requirements are predicted.¹⁶

⁸ Tajikistan is 79th out of 182 countries (in ascending vulnerability). See: <http://index.gain.org/ranking/vulnerability>

⁹ The OFDA/CRED - International Disaster Database, http://emdat.be/emdat_db/

¹⁰ Punkari et al. 2014. Climate Change and Sustainable Water Management in Central Asia. ADB Central and West Asia Working Paper Series No. 5. Asian Development Bank. Manila.

¹¹ ADB, 2013. Building Climate Resilience in the Pyanj River Basin. Technical Assistance Final Report (TA7970-TAJ). Manila.

¹² World Food Programme. 2013. *Food Security Classification Overview–June 2013*. Dushanbe.

¹³ World Bank. 2013. *Tajikistan-Reinvigorating Growth in the Khatlon Oblast*. Washington, DC.

¹⁴ ADB, 2011. *Climate Resiliency for Natural Resources Investments*. TA Consultant Final Report. Manila.

¹⁵ World Bank, 2017. *Strengthening Critical Infrastructure Against Natural Hazards Project*. Social Assessment Report. Washington DC.

¹⁶ See Footnote 10

Addressing the current and future climate-related extreme weather risks is a concern and priority at high levels of the government. This is reflected in the Tajikistan Nationally Determined Contribution submitted for the UNFCCC Paris Agreement, which prioritizes reducing the vulnerability to climate change in key sectors such as agriculture and water resources through among others improvement and modernization of hydrometeorological services, disaster risk reduction, and improved knowledge sharing.¹⁷ These adaptation priorities were likewise reflected in investments proposed during consultation among government and civil society stakeholders for improvements in water resources management including flood risk reduction in the PRB and improved knowledge sharing¹⁸ as well as the Tajikistan National Climate Change Adaptation Strategy due for approval in Q1 2018.¹⁹

C.2. Project / Programme Objective against Baseline

Describe the baseline scenario (i.e. emissions baseline, climate vulnerability baseline, key barriers, challenges and/or policies) and the outcomes and the impact that the project/programme will aim to achieve in improving the baseline scenario.

The State Agency for Hydrometeorology is Tajikistan's climate and weather agency, with a mandate to among others (i) ensure protection of vital interests of the individual, society and state from natural hazards; (ii) provide hydrometeorological information to government and civil society stakeholders; and (iii) ensure functioning of the state monitoring network and the system for collection, processing and dissemination of hydrometeorological information. This mandate includes providing warnings for extreme climate change-related weather events such as floods. However, Hydromet's capacity against the World Meteorological Organization (WMO) Strategy for Service Delivery (SSD)²⁰ and Global Framework for Climate Services (GFCS) pillars²¹ remains low. As such, Hydromet is hindered in its ability to achieve its public service mandate to inform, warn and protect the public on weather and hydrometeorological issues.

User engagement. Only 20% of surveyed stakeholders of Hydromet services expressed they were satisfied with Hydromet's forecasting services. There is a lack of two-way communication with customers to support product development and service enhancement. Hydromet lacks a framework and platform to solicit and seek feedback from users. Hydromet lacks a framework and capacity for the development and sale of fee-based services. There is an apparent lack of close relationship between Hydromet and Ministry of Emergency Situations. Advisory services are focused on general weather warnings rather than key decision-making issues.

Relevant SSD pillars: Evaluate user needs and decisions; Link service development and delivery to user needs; Share best practices and knowledge

Relevant GFCS pillars: User interface platform; Climate Service Information System

Quality of outputs and capacity development. Hydromet has a limited and patchy meteorological and hydrological monitoring network that is labor-intensive to operate. Hydromet provides local weather forecasts, for which it depends on third-party numerical weather prediction modelling and satellite imagery, and which lacks probabilistic analysis. Its weather forecasting is not integrated with analysis of hydrological response such as of flood forecasting. Its platforms and networks for rapid and broad dissemination of weather-related news and alerts to key government and civil society

¹⁷ Government of Tajikistan, 2015. *Intended Nationally Determined Contribution (INDC) towards the achievement of the global goal of the UN Framework Convention on Climate Change (UNFCCC) by the Republic of Tajikistan*. Dushanbe.

¹⁸ ADB, 2016. *Economics of Climate Change in Central and West Asia. Adaptation Component*. TA Consultant's Report. Manila.

¹⁹ ADB, 2017. *Draft National Climate Change Adaptation Strategy (under preparation)*. TA Consultant's Report. Manila.

²⁰ WMO. 2014. *The WMO Strategy for Service Delivery and its Implementation Plan*.

https://library.wmo.int/pmb_ged/wmo_1129_en.pdf

²¹ WMO. 2014. *Implementation Plan of the Global Framework for Climate Services*.

http://www.wmo.int/gfcs/sites/default/files/implementation-plan/GFCS-IMPLEMENTATION-PLAN-FINAL-14211_en.pdf

stakeholders are limited. There is a lack of motivation and integration between teams to develop new advisory and customer service skills, and apparent lack of evaluation and verification that would feed back into improved services.

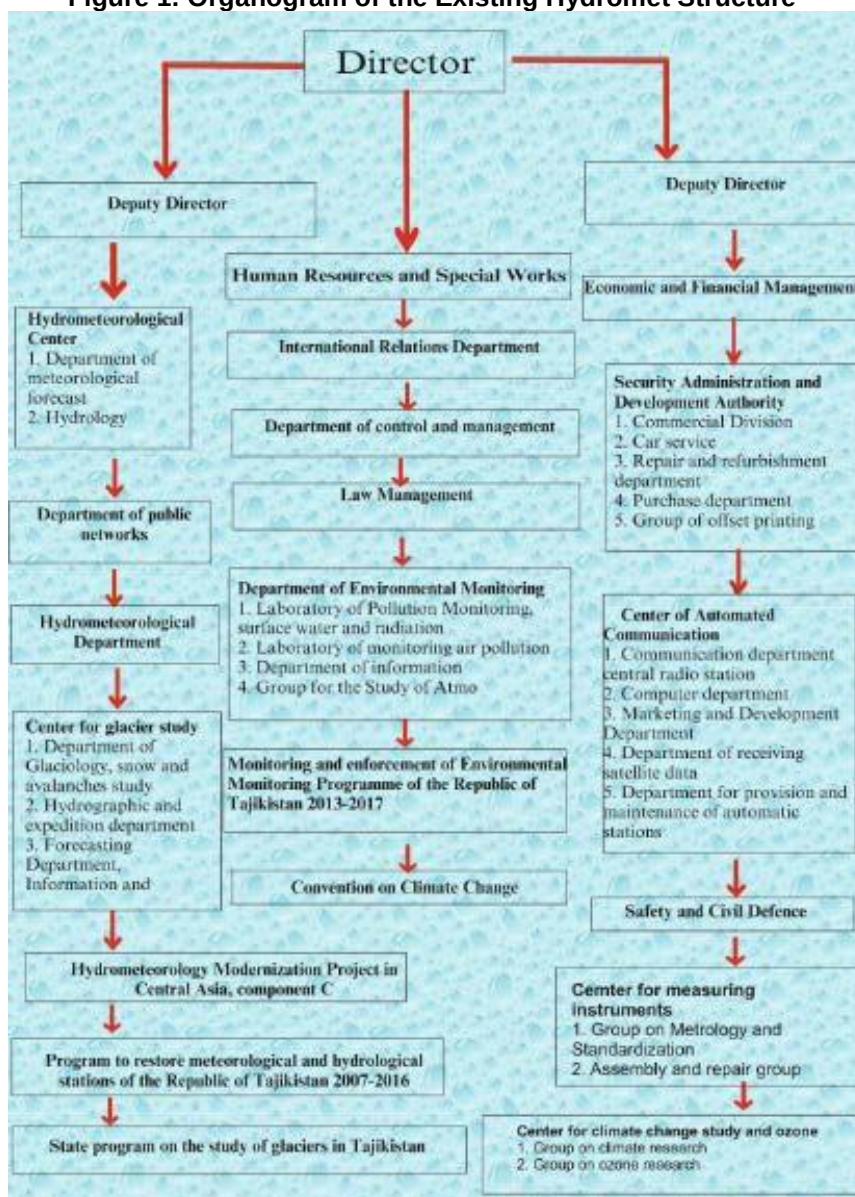
Relevant SSD pillars: Sustain improved service delivery; Develop skills needed to sustain service delivery; Evaluate and monitor service performance and outcomes

Relevant GFCS pillars: Research, Modeling and Prediction; Observations and Monitoring; Capacity Development

Underlying institutional weakness. As detailed in Attachment 1, Hydromet suffers many of the common problems facing institutions in post-Soviet countries, which cause this poor performance. It is fully financed from the state budget, but due to the country's broader budgetary constraints it receives a very limited budget in proportion to its size. Hydromet's infrastructure and buildings are poor and decaying, staff retention is low due to uncompetitive salaries, and there is limited budget for operation and maintenance (O&M).

As of March 2017, there are 708 staff in total working at Hydromet with 262 professional staff on active duty. There are 45 unfilled professional positions, roughly 15% of the total approved professional staff. Most the unfilled positions are based in Dushanbe and are currently vacant due to difficulties in the recruitment and retention of professional staff. The current organizational structure of Hydromet is complex, as illustrated in Figure 1 below. The overall leadership is at Director level, which specifically oversees several departments or units such as Human Resources, International Relations, the Department of Environmental Monitoring and the Convention on Climate Change. In addition, there are two Deputy Directors. One oversees the Hydrometeorological Centre, the Centre for Glacial Study and programs or projects such as the 'program on the study of glaciers' and 'program to restore meteorological and hydrological stations'. The second oversees the economic and financial management, automated communications, the Center for Measuring Instruments and the Centre for Climate Change and Ozone (although no work on ozone is currently performed).

Figure 1: Organogram of the Existing Hydromet Structure



Source: Hydromet

As suggested by the organogram, functional connectedness in Hydromet is lacking. A modern, effective and efficient hydromet service would have a focus on service delivery and user engagement with other programs clearly supporting the delivery of these services such as observations, communications, or cross-cutting corporate functions. Hydromet's structure, on the other hand, has had projects and programs 'bolted-on', rather than integrated into the broader organizational (and staff) structure with shared responsibilities. As such, small units or teams function independently, creating a broad structure and silos of activities. User engagement is lacking. Thus, the organization will tend to operate in silos and lacks a common vision, purpose, and close links with their (potential) customers.

Hydromet faces difficulties in recruiting and maintaining staff. For example, many of the staff in the Department of Meteorology lack professional training; instead, they learned on the job and assisted when it was operationally busy. However, many of these staff are unskilled to do the valuable and necessary improvements and assessments at times of benign conditions. Although there are a few experts within Hydromet; overall, the basic training of staff working in meteorology, hydrology, climate change impact assessments and other specialized areas is limited. Many staff at Hydromet have limited technical specialist backgrounds given the relative lack of formal training when they join

Hydromet. Individuals who do have stronger backgrounds will tend to find work in other counties or sectors where salaries are higher.

In May 2015, Hydromet's legal status was upgraded to that of a State Agency with professional staff salaries increased by roughly 40% to better align with civil service payscales. However, salaries for civil servants at Hydromet (average approximately \$150 per month) still lag behind equivalent levels in government line ministries (average \$80-\$300 per month) and are significantly lower than in the private sector (average \$100-\$3000 per month). However, in its current legal status as State Agency, Hydromet is not able to set its own staff salaries and due to limited budget availability further government-approved salary increases are unlikely to be forthcoming.

Hydromet's headquarter campus in Dushanbe comprises of a series of detached bungalows within an enclosed compound near the city center of Dushanbe. The current physical structure reflects the organogram, with activities operating out of numerous, dispersed, small offices. The buildings are in a state of disrepair with poor facilities (see Figure 2 below). The toilet facilities, particularly for women, are in very poor condition and unheated during winter months. Indeed, it is the open spaces between buildings where staff tend to congregate. Such an environment acts as a barrier to integration, innovation and efficiency. It has long been proposed that Hydromet headquarters be moved to a new site in Dushanbe. Land as already been allocated for a new Hydromet campus and construction of a 6-storey office tower at the site was initiated in the 1980s but never finished. Construction was halted and restarted at various stages, most recently in 2011 with limited annual budgetary allocations. It remains unfinished (at roughly 60% physical completion, see photo in Figure 3 below) and unable to accommodate Hydromet staff, and is likely to remain so in the coming years due to the budgetary situation of the Government of Tajikistan arising from external economic shocks.²²

Figure 2: Photos of current Hydromet headquarters buildings in Dushanbe



Source: Hydromet and Asian Development Bank

²² See: <https://www.imf.org/en/News/Articles/2017/05/30/pr17196-imf-staff-completes-2017-article-iv-mission-to-tajikistan>

Figure 3: Photo of proposed new (unfinished) Hydromet building



Source: Asian Development Bank

Ongoing projects and sustainability. Several recent and ongoing donor technical assistance and investment projects have improved Hydromet's capacity in data collection, analysis, and forecasting through technical training and modernization of equipment. The World Bank *Central Asia Hydromet Modernization Project* (CAHMP) has invested in equipment and training for to improve the meteorological and hydrological monitoring network, data storage, and short-term weather forecasting²³. The original scope of CAHMP had included the piloting of fee-based weather information products, but this was removed due to budget limitations²⁴. It is understood that additional financing is being sought for CAHMP to develop standard operating procedures and information service regulations, build capacity on additional services including glaciology and agrometeorology, further strengthen the monitoring network. The ADB *Building Capacity for Climate Resilience*²⁵ technical assistance project will invest in equipment and training for climate change scenario and impact modelling, and a climate information database. The World Bank in association with the Global Facility for Disaster Risk Reduction (GFDRR) and the World Meteorological Organization (WMO) are working with Hydromet as well as other related agencies in Central Asia to develop and implement a Central Asia Regional Flash Flood Guidance System (CAFFGS) that provides flood guidance and potentially early warnings.²⁶ The Finnish Tajikistan Meteorology Project (2014-2017) supported improvements in Hydromet's (i) capacity to provide climate services, (ii) air quality observation system, (iii) strategic and technical planning capacity, and (iv) capacity on modern observation technology and weather service process.²⁷ It is understood that a second phase will commence in 2018.

However, no donor investments have to-date sought to address the abovementioned underlying institutional barriers and weaknesses, and thus sustainability and effectiveness of these interventions is at risk. The sustainability of the benefits of training is likely to be poor with low staff retention. Purchases of equipment puts additional burden on Hydromet's already stretched O&M budget, and thus future maintenance may be inadequate. Equipment storage and operation is not sufficiently secure in Hydromet's current dilapidated campus. Indeed, ADB has postponed its climate modelling activities and procurement of high-performance computers due to the poor state of Hydromet's current campus. These will be restarted once they may be safely and securely installed in the new Hydromet campus.

As such, in the baseline, Hydromet capacity and quality of forecasting services remains low due to underlying institutional barriers and weaknesses. Hydromet is unlikely to see improvement over the current situation in the medium-term due to likely poor sustainability of ongoing donor-supported activities, and circumstances may in fact worsen as projects complete and direct support ceases.

C.3. Project / Programme Description

Describe the main activities and the planned measures of the project/programme according to each of its components.

Provide information on how the activities are linked to objectives, outputs and outcomes that the project/programme intends to achieve. The objectives, outputs and outcomes should be consistent with the information reported in the logic framework in section H.

The project will address the key barriers and weaknesses identified in Section C.2. and support the development of Hydromet to a sustainable and well-resourced institution that provides timely and accurate forecasting services. The project focuses on addressing (i) underlying institutional barriers and weaknesses of Hydromet that hamper its institutional capacity and development; and (ii) support improved capacity in the production and dissemination of forecasting services particularly for the PRB area. Unless the institutional barriers and weaknesses are addressed, it is not likely that support for Hydromet's technical capacity (by this or any other project) can be sustainable.

The project will (i) modernize Hydromet's campus and associated facilities [Component A], (ii) support legal and organizational transformation of Hydromet into a government entity with flexibility to set staff salaries and retain additional entrepreneurial revenue [Component B], (iii) undertake capacity building for improve forecasting and warning of extreme weather events [Component C], and (iv) support development and implementation of a viable business model through marketing of fee-based services [Component D]. In doing so, the project is directly and indirectly supporting Hydromet's performance against the key WMO SSD and GCFS pillars.

Component A: Hydromet campus modernization.

The project will support the modernization of the new Hydromet headquarters campus including main office building, ancillary buildings, and two mixed-use buildings. The modernization of the Hydromet campus underpins the project's contribution to the WMO SSD and GCFS pillars and achievement of the project objective of a sustainable and well-resourced Hydromet, which under this project and through a legal transformation will become a semi-autonomous entity under the Government.

Activity 1.1 Design and supervision services engaged for detailed design and system integration, preparation of bidding documents for the campus modernization. 1.2 Core operational systems modernized. The project will modernize and fit out the currently unfinished office tower described in Section C.2. and Hydromet headquarters will be relocated to this secure and modernized building (see Figure 4 for drawing). The project will install a core office IT network with high speed internet connection to be designed during implementation for integration with the World Bank CAHMP monitoring and forecasting systems and ADB-supported climate modelling facility and climate data management system and alignment with WMO guidelines²⁸. Ancillary buildings will be constructed comprising a laboratory, conference hall, and visitor's information center (see schematic of campus in Figure 5). The laboratory will be used for Hydromet's environmental testing activities. The conference hall will be used for internal activities and provide a source of revenue through rental of facilities for external engagements. The visitor's information center will be used for Hydromet's public outreach on weather and climate change education and information. Containerized offices will be installed on the campus to accommodate staff overflow.

1.3 Mixed-use buildings constructed. Two mixed-use buildings will be constructed to provide low-cost housing for Hydromet staff. **GCF finance will not contribute to this activity**. The housing facilities will be provided as an in-kind benefit to compensate for low staff salaries and ensure that Hydromet staff can remain near the office during critical extreme weather events. The lower floor of the mixed-use buildings will be rented out on a commercial basis for additional revenue for Hydromet. A modern campus will (i) allow equipment to be safely and securely operated; (ii) provide additional entrepreneurial income through rental of facilities; (iii) improved staff retention with more attractive

²³ World Bank, 2011. *Central Asia Hydromet Modernization Project*. Project Appraisal Document. Washington D.C.

²⁴ As reported by Hydromet Deputy Director and World Bank team leader in personal communication.

²⁵ ADB, 2012. *Building Capacity for Climate Resilience*. Technical Assistance Report. Manila.

²⁶ WMO. 2017. Central Asia Region Flash Flood Guidance (CARFFG) System.
<http://www.wmo.int/pages/prog/hwrf/flood/ffgs/carffg/carffg.php>

²⁷ <http://en.ilmatiiteenlaitos.fi/documents/30106/42393/tajikistan.pdf>

²⁸ WMO, 2015. *Manual on the WMO Information System*. WMO-No. 1060. Geneva.

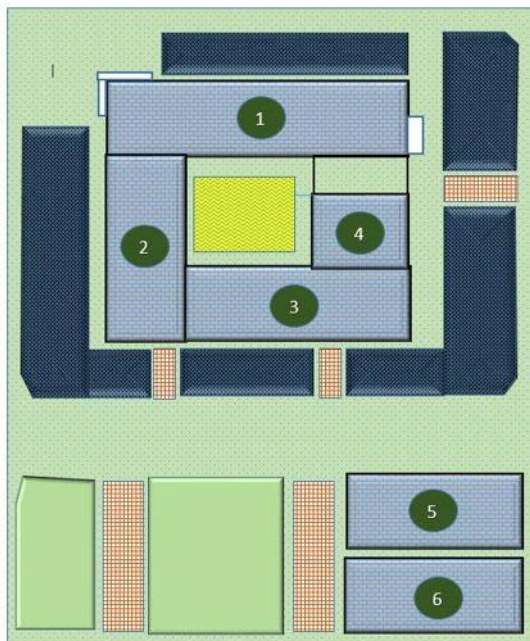
salary packages through higher salaries and in-kind benefit of low-cost housing. Higher staff retention and improved facilities, combined with the project's capacity building on management, marketing, and forecasting services, will help Hydromet improve the quality and relevance of forecasting outputs. This, in turn, will support the sale of fee-based information services and consulting services. See further analysis of options considered for Hydromet campus modernization during project preparation in Section E.6.3.

Figure 4: Architect's drawing of proposed new Hydromet building



Source: Hydromet

Figure 5: Schematic of proposed new Hydromet campus



Note: (1) Office Building, (2) Laboratory, (3) Conference Center, (4) Visitor's Information Center, (5-6) Mixed-use buildings.

Source: Hydromet

Component B: Legal and organizational transformation.

The project will address key legal and institutional barriers and weaknesses faced by Hydromet, and provide a solid operational foundation upon which it may develop and produce high quality services.

1.4 Legal reforms implemented. Hydromet's legal status will be amended to allow for more flexible and autonomous operation including internal reorganization, setting of staff salaries, and seeking and retaining additional entrepreneurial income. As described further in Section F.2., it is proposed that through a staged approach Hydromet is to be converted to full State Institution status. Hydromet will be overseen initially by a Project Steering Committee to be established under the project. Following a government amendment to Hydromet's legal status and update to the Hydromet charter, the Project Steering Committee will be formalized as a Board of Directors.

1.5 Institutional framework and quality management system enhanced. The project will enhance Hydromet's frameworks and systems for strategic planning, management, administration, and control towards ensuring output and development of quality climate services. Detailed program design will be prepared based on a gap analysis and technical needs assessment in line with WMO guidelines.²⁹ Hydromet's organizational business model and strategic planning will be updated. A viable business and marketing strategy including sustainable O&M plan will be developed to provide a framework for Hydromet to seek and retain additional entrepreneurial revenue as will be allowed under Hydromet's new legal status. Hydromet will remain primarily supported through core government funding, and thus the staging of the legal transition will be linked to the establishment of additional source of entrepreneurial income. New sources of revenue for Hydromet are expected to include (i) rental of facilities from the new Hydromet campus described in the succeeding paragraphs; (ii) sale of weather information products described under Output 3 below; and (iii) in future, consultancy and other expert services. Additional entrepreneurial income is expected to contribute more than 10% of Hydromet's annual budget by 2025 compared to 0% in 2017 (see analysis in Attachment 8). Staff structure and responsibilities will be streamlined and consolidated with formal terms of reference including quality management responsibilities and career pathways. Management, administration, and control manuals will be updated or established including sustainable O&M plan. A user feedback and service quality M&E system will be established to support reporting and stakeholder engagement on among others service experience and content, forecast timeliness and accuracy, and mediums of dissemination. Annual stakeholder opinion surveys will be undertaken on Hydromet services. A time-bound action plan will be developed and implemented for ISO 9001 certification of Hydromet's quality management system.

1.6 Capacity building and knowledge-sharing program implemented. A training program for senior and mid-level Hydromet professional staff will be designed and implemented on management and administration including strategic planning, human resource management, financial management, and career development including for women staff. Training will be conducted through a combination of workshops and seminars, and "on the job" training under Activity 1.4. Regional and international knowledge partnerships will be established, to support continued development of Hydromet capacity. The project will support a regional knowledge sharing event such as workshop, conference, or study tour. An annual stakeholder workshop will be held in-country to present project progress and plans.

Component C: Forecasting and warning of extreme weather

The project will develop Hydromet capacity riverine flood forecasting and local flood alerts in the PRB area.

1.7 Community awareness and preparedness enhanced. Gender-sensitive consultation with national and local government and civil society stakeholders will be conducted to inform (i) the selection of pilot areas and (ii) the design of flood forecasting dissemination and alerts systems including method of delivery. During and following the establishment of the flood forecasting system, the project will conduct training for stakeholders in pilot areas on flood risks and the use of flood forecasts and disaster preparedness.

1.8 Flood forecasting system developed and capacity building undertaken. Hydromet's hydrological monitoring and forecasting capacity will be developed for three (3) pilot areas in the PRB to be identified during project implementation from an indicative short-list³⁰ based on flood risk mapping and prioritization based on key risk and socio-economic

indicators and appropriate terrain. The project will install river gauges at pilot areas, and install a (likely cloud-based) flood early warning system platform in Hydromet incorporating rainfall-runoff, routing, and hydrodynamic models.

Component D: Marketing of fee-based services

It is intended that the sale of weather forecast information products will become a key source of additional entrepreneurial revenue for Hydromet in future. A system and platform for the sale of weather information and forecasting products will be developed by Hydromet and least one new weather information product will be marketed among key stakeholders in the PRB to support agricultural production and water resource management. A detailed market research analysis has been performed during project preparation to identify key issues and potential clients of Hydromet's information products (see Section F.2.).

1.9 Community awareness and weather product accessibility enhanced. Gender-sensitive consultations with farmers and stakeholder organizations in the PRB will be undertaken to inform the design of the marketing and subscription platform and prospective new weather information products. During and following the establishment of the system and new products, further gender-sensitive training and awareness campaigns will be conducted on the receipt and use of weather services.

1.10 Subscription platform and new weather products developed

A platform will be developed for managing the subscription and dissemination of weather information products, which are expected to be accessed and transmitted through web, email, and SMS. Hydromet's main website will be accordingly updated and modernized. Capacity building of Hydromet staff will be conducted on marketing and product development to support the sustainability and quality of weather product development.

The project has the following output indicators:

- Hydromet headquarters offices relocated to modernized campus. (2017 baseline: not applicable).
- Hydromet Board of Directors established. (2017 baseline: not applicable).
- Local flood forecasts issued for pilot PRB districts on Hydromet website at least 36 hours in advance with >40% probability of detection (POD) and <25% false alarm rate (FAR) (2017 baseline: not applicable)
- At least 500 individuals (of which 30% are female) are trained on flood warnings and preparedness in the PRB. (2017 baseline: 0).
- IT platform for sale of information and forecasting products established (2017 baseline: not applicable).
- At least one new weather alert product marketed to customers in the PRB by Hydromet (2017 baseline: not applicable).
- At least 500 individuals (of which 30% are female) are trained on the receipt and use of weather forecasting products in the PRB. (2017 baseline: 0).

The project has the following outcome indicators:

- Share of surveyed government and civil society stakeholders reported 'satisfied' with Hydromet's forecasting services increased to 30% (2017 baseline: 20%).
- Share of annual Hydromet budget contributed by income from sale of fee-based services increased to 1.5% (2017 baseline: 0%)

²⁹ WMO, 2013. *Guide to the Implementation of a Quality Management System for National Meteorological and Hydrological Services*. WMO-No. 1100. Geneva.

³⁰ Bichkharv (Vanj District); Ishkashim (Ishkashim District); Khirmandjo (Shuroobod District); Khorog (Shughnon District); Rushon and Shidz (Rushon District)

Equipment to be purchased

Equipment comprises approximately 10% of the base project cost (see Table 3). Identification of equipment needs has sought to (and subsequent detailed design during project implementation will) address the objectives of (i) enhancing services of Hydromet and meet the needs of stakeholders and (ii) ensuring sustainable O&M and equipment performance and lifetime. Project budget has been allocated for running costs including software licenses and subscriptions, developer services, and maintenance. Where feasible, autonomous, wireless, and cloud-based systems will be employed. It is expected that equipment purchases and estimated costs will include:

- Furniture and office equipment for campus fit-out
- IT systems including CPU, storage, backup, and wifi
- Satellite receiver system
- Automated river gauges and weather stations including telemetry
- Flood forecasting and early warning system
- Containerized offices
- Portable telemetry and monitoring equipment for field work

Contribution to WMO SSD and GFCS

The project will **directly** contribute particularly to the following key elements:

- SSD: Evaluate user needs and decisions; Evaluate and monitor service performance and outcomes; Link service development and delivery to user needs; Sustain improved service delivery; Develop skills needed to sustain service delivery
- GFCS: User interface platform; Climate Service Information System; Research, Modeling and Prediction; Capacity Development

By supporting and facilitating the sustainability of other donor interventions, notably the World Bank CAHMP project, the project will also **indirectly** contribute to the abovementioned elements as well as:

- GFCS: Observations and Monitoring

Linkages to ongoing ADB projects in Pyanj River Basin

The project's benefits and activities have strong links to two ongoing ADB water resources sector projects in the Pyanj River Basin. The *Building Climate Resilience in the Pyanj River Basin* (BCRPRB) project³¹, which runs from 2013-2020, invests in climate-proofed flood protection, irrigation, and water supply infrastructure. The project also establishes local disaster risk management committees (DRMCs) and water user associations (WUAs) to support improved disaster preparedness and response and water resources management. The *Water Resources Management in the Pyanj River Basin* (WRMPRB) project³², which runs from 2017-2022, will improve the government's WRM capacity in PRB nationally and regionally through the establishment of (i) Tajikistan-Afghanistan Joint PRB Commission (JPRBC) to support improved hydrological monitoring and (ii) river basin organizations (RBOs) in Kulob and Khorog cities for resource planning, monitoring, and development. The project will invest in a selected irrigation system, the Chubek Irrigation System in Hamadoni district (see further information in Attachment 14).

The subject proposed Hydromet project will enhance and sustain the BCRPRB and WRMPRB project's agricultural productivity and climate resilience benefits through the provision of higher-quality forecasting services for PRB beneficiaries and stakeholder institutions. Consultation and training under Components B, C and D will seek to involve the entities established under these respective projects, including the DRMCs, RBOs, WUAs, and JPRBC, to leverage the institutional structures and capacity engaged in WRM and disaster management in the region. To enhance these linkages, the subject project will be presented to the ADB Board for approval as an additional financing extension project to the WRMPRB project.

C.4. Background Information on Project / Programme Sponsor (Executing Entity)

Describe the quality of the management team, overall strategy and financial profile of the Sponsor (Executing Entity) and how it will support the project/programme in terms of equity investment, management, operations, production and marketing.

ADB will conclude a financing agreement with the Republic of Tajikistan represented by the Ministry of Finance as Executing Entity. The Ministry of Finance will cause the project to be implemented through the State Agency for Hydrometeorology (Hydromet). A project management office (PMO) will be established within Hydromet to be responsible for project implementation. The Project will also recruit consultants to support project implementation including capacity building, procurement, construction supervision, and safeguards.

Tajikistan became a member country of ADB in 1998, and since then cumulative lending, grant, and technical assistance to the country for Agriculture, Natural Resources, and Rural Development has totalled approximately \$170 million and for Public Sector Management has totalled approximately \$200 million. These represent roughly 11% and 14% of ADB's approvals in the country to date. ADB has longstanding experience in Tajikistan in supporting climate change and disaster resilience, notably in the Pyanj River basin. In addition to the abovementioned WRMPRB and BCRPRB projects, under the Tajikistan Pilot Program for Climate Resilience funded by the Climate Investment Funds, ADB is currently administering one technical assistance ("Building Capacity for Climate Resilience"). Hydromet is an executing agency (in ADB terminology) for this existing ADB technical assistance, which is developing Hydromet's capacity for modelling climate change and impacts in the country.

Hydromet is a State Agency with approximately 800 staff including approximately 300 professional staff. In addition to the abovementioned ADB technical assistance, Hydromet is implementing the World Bank Central Asian Hydromet Modernization Project to improve Hydromet's monitoring and weather forecasting capacity. A Project Management Unit (PMU) has been established within Hydromet, with responsibility for implementing the World Bank project, however, it is currently a limited team for the final stages of the project. Therefore, a new PMO will be established for the proposed project (see Section C.7 below). ADB's financial management and internal control policies and procedures will be followed, including but not limited to those described in ADB's *Loan Disbursement Handbook* (2017, as amended from time to time).

C.5. Market Overview (if applicable)

Describe the market for the product(s) or services including the historical data and forecasts.

Describe the competitive environment including the list of competitors with market shares and customer base and key differentiating factors (if applicable).

Provide pricing structures, price controls, subsidies available and government involvement (if any).

Hydromet is the government agency providing climate and weather information and forecasting to the general public in Tajikistan. There are no direct competing agencies, although private company State Unitary Enterprise 'Tajikairnavigation' provides weather services for aviation transport only in Tajikistan. A summary of Hydromet's mandate and current outputs is provided in Sections C.2. and F.2. A market research survey has been undertaken during project preparation, to assess the use and opinion in Hydromet's outputs from key government and civil society stakeholders. There is broad support for improvements in Hydromet's services, with currently only 20% respondents indicating they are 'satisfied' with current provision. Approximately 50% of respondents indicated their organizations may be willing to pay for Hydromet information and forecasts in future, particularly short- and medium-term forecasts and hydrological information. Further description of the market research is found in Section F.2.

C.6. Regulation, Taxation and Insurance (if applicable)

³¹ ADB, 2013. *Building Climate Resilience in the Pyanj River Basin*. Report and Recommendation of the President. Manila.

³² ADB, 2016. *Water Resources Management in the Pyanj River Basin Project (L3434-TAJ)*. Report and Recommendations of the President. Manila.

Provide details of government licenses or permits required for implementing and operating the project/programme, the issuing authority, and the date of issue or expected date of issue.

Describe applicable taxes and foreign exchange regulations.

Provide details on insurance policies related to project/programme.

ADB projects in Tajikistan are exempt from duties and taxes. There are no regulations affecting the import or use of foreign currency under this project.

Land use and building permits are required. Structural integrity audit has been undertaken to confirm applicable permits and conformation with local laws and structural integrity of existing buildings in the new Hydromet campus including seismic strength (see Attachment 3).

Consulting firms and contractors to be contracted by Hydromet under the project will be obliged to take the necessary insurance policies as required by the relevant government laws and/or policies. For civil works, the Contractor shall provide insurance cover from the start date to the end of the Defects Liability Period, in the amounts and deductibles stated in the particular conditions of contract for the following events, which are due to the Contractor's risks: (a) Loss of or damage to the Works, Plant, and Materials; (b) Loss of or damage to Equipment; (c) Loss of or damage to property (except the Works, Plant, Materials, and Equipment) in connection with the Contract; and (d) personal injury or death.

The project will comply with all national laws and regulations. All ADB's fiduciary, environmental and social standards, as well as ADB's procurement guidelines and policies including the ADB Procurement Guidelines (2016).

C.7. Institutional / Implementation Arrangements

Please describe in detail the governance structure of the project/programme, including but not limited to the organization structure, roles and responsibilities of the project/programme management unit, steering committee, executing entities and so on, as well as the flow of funds structure. Also describe which of these structures are already in place and which are still pending. For the pending ones, please specify the requirements to establish them.

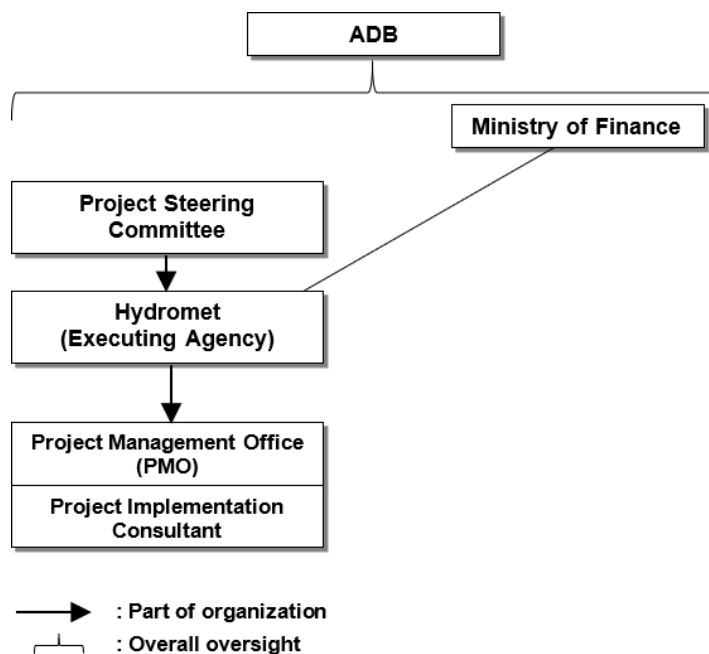
Describe construction and supervision methodology with key contractual agreements.

Describe operational arrangements with key contractual agreements following the completion of construction. If applicable, provide the credit analysis of key counterparties of key contractual agreements and/or structural mitigants to cover the counterparty risks.

The project organizational structure is presented below in Figure 6. A project steering committee (PSC) will be established comprising key stakeholders to provide oversight and advise on project implementation. Upon legal transformation of Hydromet, the PSC will serve as the Hydromet Board of Directors. A new PMO will be established within Hydromet to manage the project including finance, procurement, construction supervision, and monitoring and evaluation. A PIC will be recruited to support the Hydromet PMO in all implementation matters.

For implementation of the abovementioned activities, GCF will transfer USD 5.0 million to the GCF Account at ADB, in its capacity as Accredited Entity. As trustee of the GCF Account, ADB will make the GCF Proceeds, in the form of a grant, available to the Executing Entity in accordance with terms and conditions consistent with the Funded Activity Agreement. ADB will conclude a financing agreement with the Republic of Tajikistan, represented by the Ministry of Finance, on the management and use of the proceeds. All disbursements under the loan will be carried out in accordance with ADB's Loan Disbursement Handbook (2017, as amended from time to time). The Hydromet PMO will receive and verify claims for payments from contractors, suppliers and service providers, and consolidate and approve the requests for payment and will forward request for disbursement to ADB. ADB will pay the key suppliers, service providers and contractors via direct payment. An imprest account will be established in a local commercial bank acceptable to ADB for smaller items of expenditure of the PMO and the payment of PMO staff salaries and office operating costs. ADB, in its role as the AE, has overall responsibility and oversight for the project in line with its AMA. See further details in Attachments 1, 2, and 4.

Figure 6: Project organizational structure



Source: Asian Development Bank

Modernization of new Hydromet campus will be carried out by a construction company selected through national competitive bidding following ADB's *Procurement Guidelines* (2015, as amended from time to time). When the Project is completed, the infrastructure will be handed over to the Government of Tajikistan, per standard contract procedures.

C.8. Timetable of Project/Programme Implementation

Indicative timetable attached in Section I.

D.1. Value Added for GCF Involvement

Please specify why the GCF involvement is critical for the project/programme, in consideration of other alternatives.

In supporting the institutional transformation of Hydromet, GCF financing will provide significant value-added above the baseline case described in Section C.2 above. As indicated in B.1, GCF financing is requested to match ADB grant financing 1:1 for each expenditure category and project component, including civil works. Without the GCF financing, and thus without this proposed project, the capacity of Hydromet and the quality of its outputs will languish for the foreseeable future. The project will help overcome the key underlying institutional weaknesses and barriers of Hydromet, and ensure the sustainability and viability of Hydromet. The project will support development of weather information products and flood forecasting for residents of the Pyanj River Basin, and facilitate the achievement and sustainability of other donor interventions with Hydromet including the World Bank CAHMP project.

D.2. Exit Strategy

Please explain how the project/programme sustainability will be ensured in the long run, after the project/programme is implemented with support from the GCF and other sources, taking into consideration the long-term financial viability demonstrated in [E.6.3](#). This should include a description of strategies for longer term maintenance of physical assets (if applicable).

The project's exit strategy is to ensure that Hydromet's underlying organizational weaknesses are addressed, and the agency is on a growth trajectory towards increased self-reliance, higher technical capacity, staff retention, and high-quality outputs.

Governance

The institutional structure of Hydromet will be reformed to establish legal standing in order to flexibly set salaries, and seek and retain additional income under the Government of Tajikistan. A Board of Directors will be established to oversee management of Hydromet (see Attachment 2)

Financial and operational stability

The project will support the development and implementation of a viable business plan including marketing strategy. The institutional reform will include a reorganization of Hydromet for more efficient, effective, and capable management and output. Hydromet, as a public institution, is expected to rely on core government funding. Continued sufficient core government funding for Hydromet will be agreed between the Government of Tajikistan and ADB through the project period in line with the sustainable O&M plan, and continued monitoring by ADB after the project period will be agreed.

Maintenance

The project will develop a viable business and marketing strategy including sustainable O&M. The costs for this are included in the financial assessment (see Attachment 8).

Capacity building

A full training needs assessment and capacity building of Hydromet staff will be implemented during the project for management and organization, marketing and business strategies, and flood forecasting.

In this section, the accredited entity is expected to provide a brief description of the expected performance of the proposed project/programme against each of the Fund's six investment criteria. Activity-specific sub-criteria and indicative assessment factors, which can be found in the Fund's [Investment Framework](#), should be addressed where relevant and applicable. This section should tie into any request for concessionality made in [section B.2](#).

E.1. Impact Potential			
Potential of the project/programme to contribute to the achievement of the Fund's objectives and result areas			
E.1.1. Mitigation / adaptation impact potential			
The project will support adaptation to climate change in Tajikistan (i) directly through improvements in flood forecasting and weather products for approximately 300,000 residents (of which 150,000 are women) of the PRB and (ii) indirectly through a strengthened Hydromet agency providing improved and widely disseminated forecasting services to approximately 8.7 million³³ residents across the entire country. The benefits of the project are expected to comprise (i) avoidance adverse effects to infrastructure and livelihoods; (ii) improved climate-related data and analysis to support broader adaptation efforts including operation and maintenance, planning and investments; and (iii) leveraging and enhancing the sustainability of other donor investments in Hydromet including ongoing projects by the World Bank (to support weather monitoring and forecasting) and ADB (long-term climate forecasting), which are currently at risk due to institutional barriers and weaknesses.			
E.1.2. Key impact potential indicator			
Provide specific numerical values for the indicators below.			
GCF core indicators	Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (Mitigation only)	Annual	N/A
		Lifetime	N/A
	- Expected total number of direct and indirect beneficiaries, disaggregated by gender (reduced vulnerability or increased resilience); - Number of beneficiaries relative to total population, disaggregated by gender (adaptation only)	Total	8,735,000 (of which 4,346,000 women)
		Percentage (%)	100%
Other relevant indicators	<u>Entrepreneurial income contribution of Hydromet budget.</u> The baseline (2017) is 0% of Hydromet's annual budget is provided by income from additional entrepreneurial activities. By 2025, this is targeted to be 10%, with the remainder arising from core government funding.		
Describe the detailed methodology used for calculating the indicators above.			

³³ 2016 estimate. Source: World Bank.

Describe how the project/programme's indicator values compare to the appropriate benchmarks (i.e. the indicator values for a similar project/programme in a comparable context).

Methodology

Hydromet is a public service agency with a remit to providing national and local weather and climate information. It has a mandate to (i) ensure protection of vital interests of the individual, society and state from hazardous effects, (ii) satisfy the needs of public authorities, sectors of the economy and population for hydrometeorological information as well as information about the state of the environment and its pollution, and (iii) ensure functioning of the state monitoring network and the system for collection, processing and dissemination of hydrometeorological information. This includes providing warnings for extreme climate change-related weather events, as well as providing input to planning and development. In improving the agency, all residents of Tajikistan will be directly or indirectly benefiting.

It is estimated that roughly one quarter of the 1.3 million population of the PRB will benefit directly from the project, comprising those who may subscribe to or otherwise make use of improved weather information products and those in flood-prone areas covered by flood forecasting (to be determined during implementation). The national-level male to female population ratio is used to estimate the number of female beneficiaries.

E.2. Paradigm Shift Potential

Degree to which the proposed activity can catalyze impact beyond a one-off project/programme investment

E.2.1. Potential for scaling up and replication (Provide a numerical multiple and supporting rationale)

Describe how the proposed project/programme's expected contributions to global low-carbon and/or climate-resilient development pathways could be scaled-up and replicated including a description of the steps necessary to accomplish it.

Hydromet has a broad and important mandate to provide important monitoring and forecasting services. As a public service entity, its outputs are valuable – but un-priced and undervalued. Like other government agencies in Tajikistan, suffers from limited budget, degraded infrastructure, poor staff retention and institutional capacity. Without addressing these underlying weaknesses, there are limited prospects for significant improvement in the medium term, even with ongoing donor-supported interventions of training and equipment purchase.

The project will among others support (i) the modernization of the new Hydromet campus to ensure safe and secure facilities, and (ii) the legal transformation of Hydromet to an entity able to seek and retain additional entrepreneurial income to supplement core government funding and flexibly set staff salaries. It is expected that the key sources of income will be (i) sale of fee-based information and forecasting products; (ii) rental of facilities such as conference rooms and office space; and (iii) consulting services. The project will support the development and implementation of a viable business and marketing strategy for Hydromet and the development of at least one new weather product targeted at customers in the PRB.

This institutional transformation of Hydromet will represent a significant paradigm shift in the funding and provision of hydrometeorological services in Tajikistan and Central Asia. It is novel for a country with a substantial legacy of Soviet-era institutions and weak investment climate for entrepreneurs. The model is replicable in other hydrometeorological agencies in other countries in Central Asia. The project will help enable the scaling up of Hydromet's products and services into for example agrometeorology.

E.2.2. Potential for knowledge and learning

Describe how the project/programme contributes to the creation or strengthening of knowledge, collective learning processes, or institutions.

As described in Section C.3., the project will support the strengthening of institutional capacity of Hydromet and legal transition. The project will support improved monitoring and forecasting outputs and dissemination to the general public and its use in disaster preparedness. The institutional transformation model may be applicable to similar agencies within Tajikistan and the region. A monitoring and evaluation systems will be established in Hydromet for service development and improvement, and develop regional partnerships for knowledge development and cooperation. Hydromet will host a regional knowledge sharing event to share developments, experiences, and lessons-learned.

E.2.3. Contribution to the creation of an enabling environment

Describe how proposed measures will create conditions that are conducive to effective and sustained participation of private and public sector actors in low-carbon and/or resilient development that go beyond the program.

Describe how the proposal contributes to innovation, market development and transformation. Examples include:

- *Introducing and demonstrating a new market or a new technology in a country or a region*
- *Using innovative funding scheme such as initial public offerings and/or bond markets for projects/programme*

As described in Section C.2., insufficiently timely and accurate forecasting services for climate-related extreme weather events remain a limiting factor for efforts to reduce the vulnerability of communities in the PRB. The project will support improved forecasting services from Hydromet, to improve the disaster preparedness and response of the most vulnerable communities in Tajikistan. By supporting the provision of more timely and accurate information to the project beneficiaries, the project will empower communities to make decisions to reduce their own vulnerabilities and exposure to climate-related risks. The currently low satisfaction among key stakeholders of Hydromet's forecasting services will be increased.

As described in Section C.3., the project will support the development and implementation of a viable business model and marketing strategy for Hydromet to seek additional entrepreneurial revenue. At least one new weather product will be made available under the project, and the project will support capacity building for Hydromet to develop further information and forecasting products for income generation. This may include expansion beyond the project into areas not yet covered by Hydromet such as agrometeorology.

Private sector entities are expected to be key clients for Hydromet's fee-based services, and during marketing and product development Hydromet will engage in consultation with selected potential clients to cater services towards their priorities and needs. Further opportunities for partnerships with private entities will be explored during implementation.

E.2.4. Contribution to regulatory framework and policies

Describe how the project/programme strengthens the national / local regulatory or legal frameworks to systematically drive investment in low-emission technologies or activities, promote development of additional low-emission policies, and/or improve climate-responsive planning and development.

As described in Section C.3, the project will support the legal transition of Hydromet towards increased flexibility to seek additional income, improve forecasting services, and offer new products to the most vulnerable through outreach, consultation, and marketing. A strengthened Hydromet is a key priority for Tajikistan's NDC, and underpins the achievement of adaptation and resilience goals for national planning and development. As highlighted in Section C.2, low Hydromet capacity is a limiting factor in improved water resources management particularly in the Pyanj River Basin. Without improved forecasting and information services from Hydromet, it is unlikely that climate change and disaster risk considerations are included in planning, design, and operation of key infrastructure, as prioritized in the NDC and upcoming National Climate Change Adaptation Strategy³⁴.

E.3. Sustainable Development Potential

Wider benefits and priorities

E.3.1. Environmental, social and economic co-benefits, including gender-sensitive development impact

Economic co-benefits

The economic benefit of a strengthened Hydromet is expected to total approximately \$55 million in present value terms (at 9% discount rate) comprising avoided losses from disaster impacts (see Attachment 9).

The project will also have second- or third-order indirect effects on the wider economy, as a strengthened and more sustainable Hydromet will more effectively achieve its mandate for the country and support more climate resilient planning and development. This is in part from facilitating Hydromet's broader activities including forecasting services supported by other donor projects.

Social co-benefits

The project will support improved comfort, convenience, health, and safety among vulnerable communities in the Pyanj River Basin through improved flood forecasting and weather services. Estimated avoided affected households, deaths, and homeless due to disaster events are tabulated in Attachment 9.

Environmental co-benefits

The project does not have recorded direct environmental co-benefits, but it is noted that Hydromet undertakes environmental monitoring and testing as part of its mandate, which will indirectly benefit from institutional strengthening.

E.4. Needs of the Recipient

Vulnerability and financing needs of the beneficiary country and population

E.4.1. Vulnerability of country and beneficiary groups (Adaptation only)

³⁴ ADB, 2017. Draft National Climate Change Adaptation Strategy (under preparation). TA Consultant's Report. Manila.

Describe the scale and intensity of vulnerability of the country and beneficiary groups, and elaborate how the project/programme addresses the issue (e.g. the level of exposure to climate risks for beneficiary country and groups, overall income level, etc).

As described in Section C.1. and C.2., Tajikistan is Central Asia's poorest country and among the most vulnerable countries in the region. Tajikistan is a country highly prone to climate-related extreme weather events, notably flooding, which has caused roughly 80% of disaster mortalities in the country in the period 1990-2016.³⁵ Climate change is expected to exacerbate these adverse events and their impacts. Higher temperatures and changes in precipitation patterns are expected to cause earlier and faster snowmelt and recession of glaciers and a decline in overall water availability. Water stress conditions are likely to become more common, and flooding and landslides are likely to become more frequent and damaging³⁶.

Climate vulnerability is particularly acute in the districts along the Pyanj River Basin (PRB), the primary tributary to the Amu River in the south of the country, which are among the country's poorest and comprise a wide range of geographical and climatic conditions. PRB covers the majority of Khatlon region, the province with the largest population (2.7 million) and agriculture production (e.g. 774,000 tons cereal production) in the country, it includes the most food insecure zone among country's irrigated area,³⁷ and it is the country's poorest river basin (55% poverty rate).³⁸ In Khatlon region alone between 1998-2016, there occurred annually roughly 30 natural disaster events with 4 lives lost and damage exceeding over \$3 million per year.³⁹ Women dominate lower paying jobs in the agriculture sector – thereby bearing a disproportionate burden and vulnerability to climate-related extreme weather events. However, women have less access to resources (land, credit) and fewer are in leadership roles in WUAs and other community organizations. Annual river flows are likely to increase in the glacial sub-basins between 2015 and 2065 or 2075, due to increased melt caused by higher air temperatures. The expected change in monthly flow rates due to increased rainfall and decreased snowfall may entail an increase in the magnitude and frequency of extreme flood events. A gradual shift in the river flow seasonal distribution and increase of irrigation water requirements are predicted.⁴⁰

E.4.2. Financial, economic, social and institutional needs

Describe how the project/programme addresses the following needs:

- *Economic and social development level of the country and the affected population*
- *Absence of alternative sources of financing (e.g. fiscal or balance of payment gap that prevents from addressing the needs of the country; and lack of depth and history in the local capital market)*
- *Need for strengthening institutions and implementation capacity.*

Tajikistan is a high-debt country and the government's budgetary support for Hydromet is heavily constrained. Overall institutional capacity of Hydromet is low, and most fundamentally has underlying institutional barriers and weaknesses that prevent it from achieving its mandate and hinder sustainable improvements in institutional capacity. Without the project, Hydromet is unlikely to develop and grow its capacity beyond its current situation, and recent and ongoing donor interventions are unlikely to be sustainable.

³⁵ The OFDA/CRED - International Disaster Database, http://emdat.be/emdat_db/

³⁶ Punkari et al. 2014. Climate Change and Sustainable Water Management in Central Asia. ADB Central and West Asia Working Paper Series No. 5. Asian Development Bank. Manila.

³⁷ World Food Programme. 2013. *Food Security Classification Overview–June 2013*. Dushanbe.

³⁸ World Bank. 2013. *Tajikistan-Reinvigorating Growth in the Khatlon Oblast*. Washington, DC.

³⁹ World Bank, 2017. *Strengthening Critical Infrastructure Against Natural Hazards Project*. Social Assessment Report. Washington DC.

⁴⁰ See Footnote 10

E.5. Country Ownership

Beneficiary country (ies) ownership of, and capacity to implement, a funded project or programme

E.5.1. Existence of a national climate strategy and coherence with existing plans and policies, including NAMAs, NAPAs and NAPs

Please describe how the project/programme contributes to country's identified priorities for low-emission and climate-resilient development, and the degree to which the activity is supported by a country's enabling policy and institutional framework, or includes policy or institutional changes.

Addressing the current and future climate-related extreme weather risks is a concern and priority at high levels of the government. This is reflected in the Tajikistan Nationally Determined Contribution submitted for the UNFCCC Paris Agreement, which prioritizes reducing the vulnerability to climate change in key sectors such as agriculture and water resources through among others improvement and modernization of hydrometeorological services, disaster risk reduction, and improved knowledge sharing. These adaptation priorities were likewise reflected in investments proposed during consultation among government and civil society stakeholders for improvements in water resources management including flood risk reduction in the PRB and improved knowledge sharing⁴¹ as well as the Tajikistan National Climate Change Adaptation Strategy due for approval in October 2017.⁴²

The project will contribute to international strategy and policy frameworks for development and disaster risk reduction. The project will contribute to the outcome and targets of the Sendai Framework for Disaster Risk Reduction⁴³, particularly through improving understanding of disaster risk (Priority 1) and enhancing disaster preparedness (Priority 4). The project will contribute to the Sustainable Development Goals⁴⁴ particularly Goal 13: "Take urgent action to combat climate change and its impacts" through among others improving awareness and capacity on adaptation and early warning. The project will contribute to the implementation of the Global Framework for Climate Services⁴⁵ through improving provision of climate information (Goals 1 and 2) and strengthening engagement of providers and users of climate services (Goal 4)

E.5.2. Capacity of accredited entities and executing entities to deliver

Please describe experience and track record of the accredited entity and executing entities with respect to the activities that they are expected to undertake in the proposed project/programme.

The accredited entity is the ADB. ADB is a multilateral development finance institution that provides loans, grants and technical assistance to its developing member countries. ADB is composed of 67 members, 48 of which are from the

⁴¹ ADB, 2016. Economics of Climate Change in Central and West Asia. Adaptation Component. TA Consultant's Report. Manila.

⁴² ADB, 2017. Draft National Climate Change Adaptation Strategy (under preparation). TA Consultant's Report. Manila.

⁴³ United Nations, 2015. *Sendai Framework for Disaster Risk Reduction 2015-2030*. Third UN World Conference. Sendai.

⁴⁴ United Nations, 2015. *Transforming our world: the 2030 Agenda for Sustainable Development*. Resolution adopted by the General Assembly on 25 September 2015. New York.

⁴⁵ WMO, 2014. *Implementation Plan of the Global Framework for Climate Services*. Geneva.

Asia and Pacific region. ADB's clients are its member governments, who are also its shareholders. In addition, ADB provides direct assistance to private enterprises of developing member countries through equity investments and loans. In 2016, loan, grant and technical assistance approvals to ADB's developing member countries amounted to \$17.8 billion, and total cofinancing mobilized, with donor support, amounted to \$13.9 billion, bringing total sovereign operations to \$31.7 billion in 2016. Nonsovereign operations for the same year amounted to \$2.5 billion.

Tajikistan became a member country of ADB in 1998, and since then cumulative lending, grant, and technical assistance to the country for Agriculture, Natural Resources, and Rural Development has totalled approximately \$170 million and for Public Sector Management has totalled approximately \$200 million. These represent roughly 11% and 14% of ADB's approvals in the country to date. ADB has longstanding experience in Tajikistan in supporting climate change and disaster resilience, notably in the Pyanj River basin. Under the Tajikistan Pilot Program for Climate Resilience funded by the Climate Investment Funds, ADB is currently administering one technical assistance ("Building Capacity for Climate Resilience") and one investment project ("Building Climate Resilience in the Pyanj River Basin"). Hydromet is an executing entity ("executing agency" in ADB terminology) for the existing and ongoing ADB technical assistance, which is developing Hydromet's capacity for modelling climate change and impacts in the country, including development of a climate information and data system.

The subject project's executing entity is the Ministry of Finance and will be implemented by Hydromet. See Section C.4. for information on its capacity to implement the project.

E.5.3. Engagement with NDAs, civil society organizations and other relevant stakeholders

The project was initiated by the Center of Environmental Protection (COEP) and Hydromet, who requested ADB's support in modernizing the Hydromet campus. Following an ADB Consultation Mission in March 2016, including high-level consultation with the First Deputy Prime Ministers and representatives of COEP and Hydromet, the Government requested ADB to develop the institutional transformation project for Hydromet including campus modernization. The GCF NDA for Tajikistan is the Chairman of COEP.

The ADB project team and project preparation consultants have worked closely with Hydromet management and staff to develop the project. A detailed market research survey was conducted of government and civil society stakeholders (including ALRI and MEWR) on their use, opinion, and interest in Hydromet products and services (see Attachment 1).

E.6. Efficiency and Effectiveness

Economic and, if appropriate, financial soundness of the project/programme

E.6.1. Cost-effectiveness and efficiency

Describe how the financial structure is adequate and reasonable in order to achieve the proposal's objectives, including addressing existing bottlenecks and/or barriers; providing the least concessionality; and without crowding out private and other public investment.

Please describe the efficiency and effectiveness, taking into account the total project financing and the mitigation/adaptation impact that the project/programme aims to achieve, and explain how this compares to an appropriate benchmark. For mitigation, please make a reference to [E.6.5 \(core indicator for the cost per tCO2eq\)](#).

In light of the broad and public-service nature of Hydromet, the beneficiaries of improved services will be both direct and indirect. If it is taken that all 8.7 million inhabitants can benefit from improved Hydromet services, the cost per beneficiary of the project is \$1.2.

The project economic analysis is presented in Attachment 9. As summarized in Table 5, the economic net present value (ENPV) at 9% discount rate of the project is \$47 million over 30 years against a project cost of \$11.0 million,

with an EIRR of 29%. This result is in line with the findings of the literature indicating high benefit-cost ratio for investments in weather and climate services⁴⁶. The finding is robust to alternative cost and benefit cases.

Table 4: Results of economic evaluation and sensitivity analysis

Results of Evaluation	Change	ENPV@9% (\$ million)	EIRR (%)	Switching Value
Project Case		47.1	29.0%	
Sensitivity Scenarios				
Case 1 - Increase in Capital Costs	+40%	43.5	23.9%	525%
Case 2 - Decrease in benefit	-50%	19.0	19.3%	-84%
Case 3 - Benefit delay by 2 years	-2	36.1	22.8%	-15
Case 4 - Combination of cases 1, 2 and 3	as above	9.9	13.2%	n/a

Source: Asian Development Bank

Note: ENPV@9% = economic net present value at 9% discount rate; EIRR = economic internal rate of return; Switching Value = level of sensitivity scenario at which ENPV = 0.

E.6.2. Co-financing, leveraging and mobilized long-term investments (mitigation only)

As presented in Section B.1., GCF's co-financing ratio for the project is 50%.

E.6.3. Financial viability

Please specify the expected economic and financial rate of return with and without the Fund's support, based on the analysis conducted in [E.1](#).

The project financial analysis is presented in Attachment 8. After legal transformation, it is envisaged that Hydromet will remain primarily funded by core government budget in light of its public service mandate rather than stand alone as a profit-making entity. This is consistent with other similar meteorological agencies elsewhere with budget flexibility/autonomy arrangements, such the UK Met Office⁴⁷. As such, the project financial analysis focuses on the key financial issues related to the project's contribution to a sustainable and well-resourced Hydromet agency, notably projected (i) entrepreneurial revenues; (ii) O&M costs; (iii) staff salary package; and (iv) required core government budget support for Hydromet.

Salaries for key professional civil servant staff at Hydromet (average approximately \$150 per month) lag well behind equivalent levels in government line ministries (\$80-\$300 per month) and are significantly lower than in the private sector (\$100-\$3000 per month). This constrains staff retention, recruitment, and development. However, in its current legal status as State Agency, Hydromet is not able to set its own staff salaries and due to limited budget availability further government-approved salary increases are unlikely to be forthcoming.

The modernization of the Hydromet campus underpins the achievement of the project objectives of a sustainable and

⁴⁶ World Meteorological Organization, 2015. *Valuing Weather and Climate: Economic Assessment of Meteorological and Hydrological Services*. WMO Report No. 1153. Geneva.

⁴⁷ UK Met Office, 2017. *Annual Report and Accounts 2016/2017*. Exeter.

well-resourced Hydromet that produced timely and accurate forecasts. A modern campus will (i) ensure equipment to be safely and securely operated; (ii) provide additional entrepreneurial income through rental of facilities; (iii) improve staff retention with more attractive salary packages through higher salaries and in-kind benefit of low-cost housing. Higher staff retention and improved facilities, combined with the legal transformation of Hydromet and capacity building on management, marketing, and forecasting services, will help Hydromet improve the quality and relevance of forecasting outputs. This, in turn, will support additional entrepreneurial income from the sale of fee-based forecasting services and consulting services.

Four project options for campus modernization were considered: (i) Option A – Office tower (Building 1), ancillary buildings (Buildings 2,3,4), and two mixed-use buildings (Buildings 5 and 6); (ii) Option B – Office tower (Building 1), ancillary buildings (Buildings 2,3,4), and one mixed-use building (Building 5); Option C – Office tower (Building 1) and ancillary buildings (Buildings 2,3,4) only; and (iv) Capacity Building Only scenario –project includes only capacity building for Hydromet forecasting services, the new office building (Building 1) is completed in 2023 using State funds, and no legal transformation of Hydromet is sought, as there would be limited opportunity for rental income in the absence of a modernized campus during the project implementation period. These alternative scenarios are compared against the 'Without Project' baseline scenario, wherein no Hydromet project is undertaken and new office building is completed in 2023 using State funds.

Option A was selected for the project.

The key financial indicators for 2020 and 2030 for the alternative options and baseline are presented in Table 6 below. The project's contribution to the objective of a sustainable and well-resourced Hydromet will depend on the extent to which it can address key underlying institutional weaknesses and barriers. In particular dilapidated facilities and poor staff retention and development hamper Hydromet's achievement of its key public service mandates.

The share of entrepreneurial income in Hydromet's total budget reaches roughly 10% in 2025 and 15% in 2045 under Options A, B, and C. By comparison, the UK Met Office receives approximately 15% from non-governmental sources.⁴⁸ Thus, required annual government contributions to the Hydromet budget are lower under Options A, B, and C compared to the Without Project baseline by up to 4%. However, required contributions are higher under Capacity Building Only scenario, as no additional entrepreneurial revenue is gained and capital investments increase O&M costs.

Options C and Capacity Building Only provide only limited increase in staff salary packages, and the latter also does not modernize the Hydromet campus. It is unlikely that donor investments under these scenarios are will be sustainable in the medium-term and thus will unlikely help deviate Hydromet's development and level of forecasting services beyond the current limited level. Option B provides a modernized office campus, but only a limited number of civil servants will receive a significant increase in salary package. This will constrain and potentially hamper the sustainability of the benefits of the overall investment in campus modernization and capacity building. As such, Options B, C, and Capacity Building Only were deemed insufficient and appropriate.

⁴⁸ UK Met Office, 2017. *Annual Report and Accounts 2016/2017*. Exeter.

Table 5: Summary projected financial indicators of alternative scenarios

Indicator	Without Project	Option A	Option B	Option C	Capacity Building Only
Capital Investment (\$)	1,192,602	6,024,155	3,949,248	1,874,342	1,522,602
Post-tax entrepreneurial income (\$ p.a)					
2020	0	74,283	55,083	35,883	0
2030	0	274,631	248,827	223,024	0
2040	0	520,905	486,228	451,551	0
Government contribution (\$ p.a.)					
2020	1,125,410	1,110,553	1,114,393	1,118,233	1,125,410
2030	1,784,757	1,736,648	1,741,808	1,746,969	1,788,057
2040	2,835,357	2,737,993	2,744,928	2,751,864	2,838,657
Civil servant salary package^a (\$ p.m.)					
2020	173	439	436 ^b	179	173
2030	282	631	626	319	282
2040	459	867	862	534	459
^a Includes in-kind housing benefit under Option A and B ^b Only applicable for 40 of approximately 64 civil servant positions. The remainder will have salary packages comparable to those under Option C. Source: Asian Development Bank					
E.6.4. Application of best practices					
<i>Please explain how best available technologies and practices are considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices.</i> During project preparation, the institutional assessment of Hydromet included a comparison against the WMO SSD and GFCS⁴⁹ for a comparison against best practices for climate and weather service entities. A market research survey was undertaken to understand and incorporate the needs and expectations of key stakeholders and potential clients of fee-based services. The project will support organizational and legal transformation and Hydromet to allow increased flexibility to set staff salaries and seek and retain additional revenues.					
E.6.5. Key efficiency and effectiveness indicators					
	Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation only)				

⁴⁹ WMO. 2014. The WMO Strategy for Service Delivery and its Implementation Plan. Geneva. See: https://library.wmo.int/pmb_ged/wmo_1129_en.pdf

GCF core indicators	N/A
	Expected volume of finance to be leveraged by the proposed project/programme and as a result of the Fund's financing, disaggregated by public and private sources (mitigation only)
	N/A
Other relevant indicators (e.g. estimated cost per co-benefit generated as a result of the project/programme)	

** The information can be drawn from the project/programme appraisal document.*

F.1. Economic and Financial Analysis

Please provide the narrative and rationale for the detailed economic and financial analysis (including the financial model, taking into consideration the information provided in [section E.6.3](#)).

Based on the above analysis, please provide economic and financial justification (both qualitative and quantitative) for the concessionality that GCF provides, with a reference to the financial structure proposed in section B.2.

The economic and financial analysis considers the circumstances and uncertainties of investments in climate and weather services. Improvements in weather and climate services provide benefits only through the way the services are received and used by the end user. The benefits may include avoided losses to infrastructure, livelihoods, or life; improved day-to-day convenience, safety and comfort; improved water resources quality and management; and increased productivity in transport, agriculture, energy sectors. However, identifying and quantifying all such benefits may be difficult or impossible; the unknowns and uncertainties in the "value-adding" process add complications to economic analysis of such investments.⁵⁰ Furthermore, weather and climate service agencies are typically public service institutions. They may in some cases seek and retain additional entrepreneurial income, but annual operating costs are typically fully or largely funded from the government budget.

As such, the economic and financial analysis of this project focuses on the following respective issues:

1. The sufficiency of **economic returns** to the investment project, capturing the incremental capital and O&M costs and benefits arising from improved climate services under the project. The analysis takes a conservative approach to the benefits, capturing only the expected the avoided damages from improved forecasting of disasters arising from broader long-term Hydromet strengthening from the project. Additional, uncertain and possibly unquantifiable economic, social, and environmental benefits are not included.

⁵⁰ World Meteorological Organization, 2015. *Valuing Weather and Climate: Economic Assessment of Meteorological and Hydrological Services*. WMO Report No. 1153. Geneva.

2. The long-term **financial viability** of the strengthened Hydromet considering O&M costs, prospective additional entrepreneurial income, staff retention through increased salaries, and necessary continued government core budgetary support. The project's contribution to the objective of a sustainable and well-resourced Hydromet will depend on the extent to which it can address key underlying institutional weaknesses and barriers. Dilapidated facilities and poor staff retention and development hamper Hydromet's achievement of its key public service mandates.

The results are summarized in Section E.6.1 and E.6.3, with further detail in Attachment 8 and 9.

Grant finance is requested as Hydromet is not a profit-making enterprise and is expected to continue to rely on core government funding for the foreseeable future. Tajikistan is a country with high debt levels and high vulnerability to climate change.

F.2. Technical Evaluation

Please provide an assessment from the technical perspective. If a particular technological solution has been chosen, describe why it is the most appropriate for this project/programme.

The project's rationale and interventions are described in Section C.2. and C.3, which were underpinned by institutional, market, legal, and infrastructure assessments. Summaries are provided below, with full reports in Attachments 1, 2, and 3 respectively.

Institutional development and forecasting services

A detailed institutional assessment has been undertaken of Hydromet assessing (i) current organizational structure and capacity of Hydromet to fulfill its mandate, (ii) key needs of the country, and (iii) recommended directions for development.

Hydromet's meteorological monitoring station is limited (although recently improved by World Bank's CAHMP project) and labor intensive with delays in data transmission. Hydromet do not operate their own numerical weather prediction (NWP) model for short-term weather forecasting. Instead they access NWP output using the Internet from a selection of modeling centres, such as ECMWF and Deutscher Wetterdienst where the COSMO model was developed. Hydromet are therefore reliant on the Internet to access NWP data, as well as other important information such as satellite imagery. At Hydromet the NWP data used is at a reasonably coarse resolution, and largely deterministic. There is no integration of the meteorological forecast with the predicted hydrological response. Hydrological, meteorological, agro-meteorological observations cannot be implemented due to poor functioning equipment, weak coverage due to low numbers of equipment, and lack of training. Indeed, the World Bank noted that '*hydrological data, which is critically important for the economy, is completely inadequate*⁵¹'. Significant improvements are required in Hydromet's capacity for forecasting of extreme weather and floods.

Organizationally, there is a lack of functional programs that oversee and integrate activities. Instead, the current structure of Hydromet has projects compartmentalized and 'bolted-on' to the existing structure. Second, there is an apparent absence of a clear operational focus of forecasting delivery. Third, there is very little evidence of engagement, understanding the evolving needs of the 'user' or 'customer', and product development. As of March 2017, there were 708 staff in total working at Hydromet with 262 professional staff on active duty. There are 45 unfilled professional positions, roughly 15% of the total approved professional staff. Most the unfilled positions are

⁵¹World Bank. 2011. Central Asia Hydrometeorology Modernization Project. Project Appraisal Document. Washington DC.

based in Dushanbe and are currently vacant due to difficulties in the recruitment and retention of professional staff. The majority of the unfilled positions are based in Dushanbe and are currently vacant due to difficulties in the recruitment and retention of professional staff. Hydromet faces difficulties in recruiting and maintaining staff. Many of the staff in the Department of Meteorology lack professional training; instead, they learned on the job and assisted when it was operationally busy. However, many of these staff are unskilled to do the valuable and necessary improvements and assessments at times of benign conditions.

Consultation with the stakeholders (i.e. users of Hydromet data) identified consistent views and needs for Tajikistan. They broadly signaled a lack of reliability and therefore trust in the forecasts produced by Hydromet – although stakeholders were often sympathetic to their lack of resourcing and monitoring available to Hydromet. Results from a market survey (see below) indicated that although users required a better service, roughly 70% of respondents did consider that there had been an increase in reliability of forecasts over the last 5 years. The desire for more information and awareness leading up to severe weather or flood related disasters; and better coordination of weather and flood information during an event. A lack of understanding of what historical data may be available, and what longer range weekly to seasonal information may be available. For example, whilst historical data would assist in strategic planning of hydroelectric power stations, the longer range could be used to optimize their operations. Tajikistan currently produces ‘traditional weather and hydrological information’, but there is a potentially an undeveloped weather and climate services ‘market’. Access and communication of weather and climate informations should be improved using electronic methods such as the internet and a wider range of media sources.

The intended pilots under Component C of the project may inform potential further expansion of monitoring across Tajikistan and development of hydrometeorological and flood forecasting services. Based on gaps in the current network, it is recommended that additional new automated river gauge are procured with ratings conducted and telemetry installed. It is noted that the testing of automatic gauges is ongoing at present under the World Bank CAHMP project. Additional operating and maintenance costs should be incorporated into project for initial startup. Automated and telemetered river gauges in the PRB should be configured into a Flood Early Warning system that integrates observed and forecast meteorological and hydrological data. Hydrological models can be developed using software that is freely available (such as Flood Modeller⁵²), and where hydrological models are already established these can be ported across to run on a flexible cloud-based platform. Recent advances in software design and computing have led to significant improvements that Hydromet could take advantage of. Instead of large capital investment in computing hardware a cloud-based system should be adopted. For example, the national hydrological service in Wales, National Resources Wales, have recently adopted a cloud based system to run hydrological, coastal and hydraulic routing models.⁵³

Market research

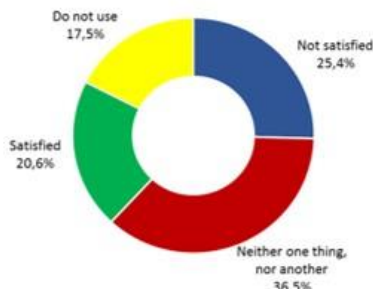
A market research study of 63 government and local stakeholders to assess their interest, opinion, and willingness to pay for Hydromet’s information and forecasting services. There was broad interest and support for Hydromet’s services, but only 20% of respondents express satisfaction with Hydromet’s forecasting services (see Figure 7). However, approximately 70% of respondents indicated that forecasts had improved in the last 5 years.

⁵² Flood Modeller <https://www.floodmodeller.com/en-gb/about/>

⁵³ Deltares – Delft-FEWS <https://www.deltares.nl/en/software/flood-forecasting-system-delft-fews-2/>

Figure 7: Respondents' assessment of satisfaction of Hydromet forecasting

*Satisfaction of Tajikhydromet
forecasting services, N=63*



Approximately 50% of respondents expressed a broad willingness for their institutions to in principle to pay for Hydromet services, while some agencies indicated that Hydromet should provide its services for free as a public entity. Among those expressing willingness to pay for Hydromet services, the prospective services of most interest were historical data on weather or river flow, (ii) short-term forecasts [up to 2 days], and (iii) mid-term forecasts [up to 7 days]. Stated willingness to pay was generally up to 100 TJS (roughly 10 USD) for each forecast. The most desired way of receiving this information is e-mail (75% of respondents), physical delivery to organization (38%), internet site (25%) and SMS notification (19%).

Legal analysis

Hydromet is a Government Agency transformed by a Government Resolution (decree) dated May 30, 2015 into a civil service organization, whose civil service rules apply to 64 staff while the majority fall outside the civil service. After May 2015 the State agency (civil service) budget from Government is separated from the rest of Hydromet staff who fall under a State institution type of structure. Hydromet is a hybrid entity applying two different sets of rules to its staff (remuneration, promotions, incentives etc.) but also enjoying certain flexible management arrangements under the law that regulates a State institution. According to law and regulations applicable to State agencies, any revenue generated by Hydromet, whether through services rendered, products and services sold or rentals from assets accrued, must under an Agency status, be "surrendered" first to MOF. Discretion to put revenues that accrue from sources outside the State budget at the disposal of Hydromet lies entirely with MOF except for funds provided to Hydromet for project implementation through sovereign loans and grants governed by project agreements.

For Hydromet to retain and manage its revenues flexibly and provide performance-based incentives to staff (civil servants and non-civil servant), current restrictions resulting from it being a Government Agency need to be addressed. The Government has assured in a letter issued by the Office of the First Deputy Prime Minister of the Government of Tajikistan of its principled support for the transformation of Hydromet into a semi-autonomous agency with flexibility towards salaries and retaining revenues. There may be a need to consider a legal status of Hydromet that is different than that of an Agency (civil service) allowing it to carry out the institutional transformation under a grant project. In this regard, the example of the National Center for Accreditation, which has gone through a process of legal transformation and is awaiting its approval as a State Institution for similar justifications may be worth considering.

Table 7 below provides an overview of the type of legal structure and under which Ministry or governing entity the Hydromet entity falls / reports to. There is no consistent or uniform pattern that can be gleaned from examining the Central Asian countries including the Russian Federation.

Table 6: Comparison of Region 1 Hydromet institutions

No.	Country	Type	Governing Body	Equivalent legal structure in Tajikistan
1	Kazakhstan ⁵⁴	State Unitary Enterprise	Ministry of Energy	State unitary enterprise
2	Kyrgyzstan ⁵⁵	Agency for Hydrometeorology	Ministry of Emergency Situations	Agency
3	Uzbekistan ⁵⁶	Center of Hydrometeorology Service	Ministry of Emergency Situations	State Institution
4	Turkmenistan ⁵⁷	National Committee for Hydrometeorology	Cabinet of Ministers	National Committee. However, in Tajikistan national committees are advisory bodies while in Turkmenistan it is close to an Agency.
5	Russian federation ⁵⁸	Federal Service for Hydrometeorology and Environmental Monitoring	Ministry of Natural Resources and Ecology	State Service having more inspection function

Source. Asian Development Bank

The following broad steps are considered for the proposed legal transformation of Hydromet, to take place in a staged approach taking into account the project progress and Hydromet capacity:

- (i) Establishment of a Project Steering Committee (PSC) to oversee project progress. The PSC will become the Supervisory Council (Board) of Hydromet upon conversion to State Institution status.
- (ii) Confirm that Hydromet continues to receive adequate State budget support and ensure that the core government support is not reduced to offset expected future increased revenue from additional entrepreneurial sources.
- (iii) Confirm that new Hydromet campus assets currently registered with COEP are transferred to Hydromet on completion of physical structures and any new structures are registered in the name of Hydromet. This will ensure that the use and benefits of the assets purchased under the project accrue to Hydromet alone.
- (iv) Change Hydromet legal status from a Government Agency to that of a State Institution governed by a Supervisory Council (Board) under the Government of Tajikistan to ensure retention of revenue accrued from rentals and service provisions and flexibilities to set staff salaries.

Infrastructure audit assessment

A structural audit of the unfinished office building was undertaken for project preparation. The recommendations will be incorporated into the design of the Hydromet modernization. The review concluded that:

[T]aking into account the elimination of the identified shortcomings, the cross-section and reinforcement of reinforced concrete structures provides the necessary load-bearing capacity of these building elements. The loaded capability and stability of the main structural elements of the building are sufficient for the perception of design loads, taking into account the 9-point earthquake action.

The review recommended minor remedial measures for the completion of the building, including:

- (i) *The Client of the Construction of object must agree on a change in the design decision on the upper part of the erected block-2 with the KVD PILT Institute "Dushanbeshahrsoz" and with the Main Department of Architecture and Urban Planning in Dushanbe;*

- (ii) *The Client of the Construction of object must conduct additional state expertise of the modified part of the project in accordance with the Urban Development Code of the Republic of Tajikistan (Chapter 9, Article 64, paragraph 3);*
- (iii) *Clean all bare reinforcing bars and fittings from rust;*
- (iv) *To strengthen the columns of the sixth floor and the technical room on the roof of the building in the axes 11-Г, 11-В, 14-Г, 14-В, 16-В, 16-Г by metal clips from corners L75x75mm, connected with each other by plates 50x5mm every 500mm;*
- (v) *To restore the separated sections of reinforced concrete structures with concrete of the appropriate grade to concrete structures; and*
- (vi) *In all the partitions in newly pierced doorways to arrange reinforced concrete or reinforced brick lintel blocks.*

F.3. Environmental, Social Assessment, including Gender Considerations

Describe the main outcome of the environment and social impact assessment. Specify the Environmental and Social Management Plan, and how the project/programme will avoid or mitigate negative impacts at each stage (e.g. preparation, implementation and operation), in accordance with the Fund's Environmental and Social Safeguard (ESS) standard. Also describe how the gender aspect is considered in accordance with the Fund's Gender Policy and Action Plan.

Environmental Considerations

The project is category B as per ADB's Safeguard Policy Statement (SPS) 2009, and in line with ADB guidelines an initial environmental evaluation (IEE) was undertaken including consultation with local and national government and civil society stakeholders. The IEE assessed the potential impacts associated with the proposed civil works (modernization of Hydromet campus): pollution from wastes, noise, dust, and air pollution, health hazards and labor safety issues. It was determined that they are all expected to be typical for small scale construction/rehabilitation works, temporary by nature and site specific and can be easily mitigated by applying best construction practices and relevant mitigation measures. An environmental management plan (EMP) has been prepared setting out environmental mitigation measures to be incorporated in the design and implementation of the project, to be implemented by the construction contractor and overseen by Hydromet. It was concluded that no further environmental assessment was necessary. An audit of the existing structure in the new Hydromet campus has been undertaken to assess structural integrity, appropriate permitting and standards (including seismic risks), and environmental issues (see Attachment 3). The draft IEE and EMP are included in the Attachment 10.

Social Considerations

This project is category C for involuntary resettlement and indigenous peoples as per the SPS. All physical civil works activities will be conducted within the land which Hydromet has already owned therefore no land acquisition and resettlement is expected in this project. The projects will not impact any indigenous peoples, as per definition of SPS and will be confirmed during implementation.

⁵⁴ Kazakhstan at: <http://www.kazhydromet.kz/ru/history>

⁵⁵ Kyrgyzstan at: <http://meteo.kg/index.php?page=2> ; <http://mes.kg/ru/about/subordinate/kyrgyzgidromet/>

⁵⁶ Uzbekistan at: <http://www.meteo.uz/>

⁵⁷ Turkmenistan at: <http://www.meteo.gov.tm/ru/>

⁵⁸ Russian Federation at: <http://www.meteorf.ru/>

During project preparation, extensive consultation has been undertaken with government and civil society stakeholders on their awareness, use, and satisfaction of Hydromet services. Reported support for improved Hydromet services is high with potentially large benefits to direct and indirect beneficiaries. During project implementation, the development of flood forecasting and weather products will be informed by gender-sensitive consultation of national and local stakeholders, including RBOs and WUAs.

Gender considerations

A poverty, social, and gender assessment was undertaken (see Attachment 13) and a gender action plan (GAP) and Summary Poverty Reduction and Social Strategy prepared (see Attachments 5 and 6).

The gender assessment identified that key gender issues are in leadership and access to resources. Within Hydromet organization, only 3 out of 19 senior level positions are held by women. Among the project's broader beneficiaries, women dominate lower paying jobs in the agriculture sector – thereby bearing a disproportionate burden and vulnerability to climate-related extreme weather events. However, women have less access to resources (land, credit) and fewer are in leadership roles in WUAs and other community organizations.

The GAP includes the following key performance targets and activities:

- ☐ An average of 25% of staff positions at Chief and Deputy Chief of Department, Chief of Groups, Heads of Units, Laboratory, and Stations are held by women. (Baseline 2017: average 20%)
- ☐ Modernized campus is designed and constructed with adequate gender-appropriate facilities (e.g. sanitary, changing rooms, breast feeding rooms, refreshment areas). (Baseline 2017: not applicable)
- ☐ At least 25 qualified female staff from Hydromet are trained in administration and management including leadership, financial management, and commercial management. (Baseline 2017: 0)
- ☐ Clear career path opportunities and career track profiles are provided to every staff member. (Baseline 2017: not applicable)
- ☐ Participants in community consultation and participation activities will comprise 30% women.

An gender specialist will be recruited under the PIC to assess, consultant, train, and support management of the implementation of the GAP. A gender specialist will be recruited under the PMO for GAP implementation.

F.4. Financial Management and Procurement

Describe the project/programme's financial management and procurement, including financial accounting, disbursement methods and auditing.

Financial management and procurement capacity assessment

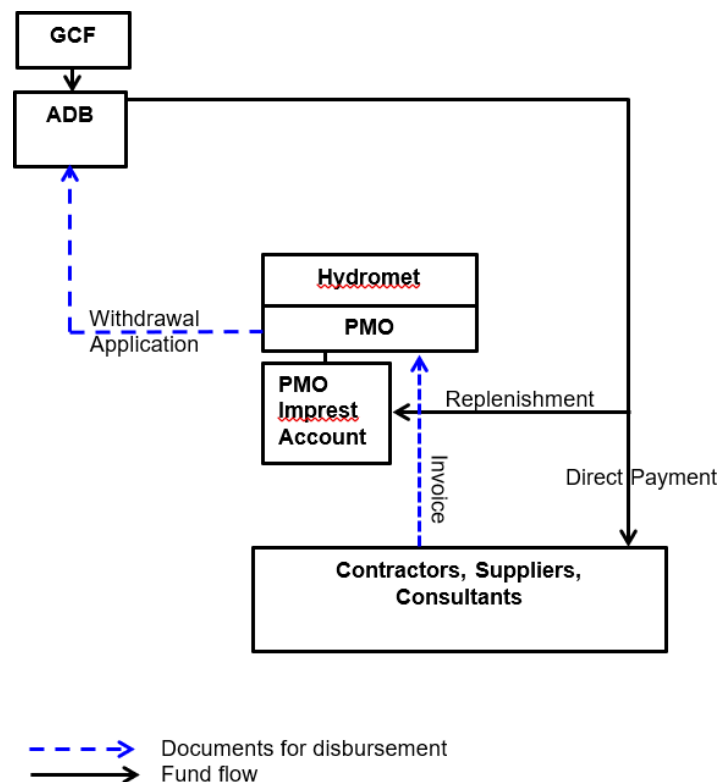
A full assessment has been undertaken of the financial management and procurement capacity of the executing and implementing entities (see Attachments 4 and 7).

The FMA for Hydromet as the implementing entity of the subject proposed project concluded that it represents a substantial risk (defined as high impact but low probability risks) due to less effective internal controls, accounting and financial reporting, and funds flow. There is currently no PMU within Hydromet that can manage the project, and as described above in Section C.4. a new PMO will need to be established for the project. The FMA recommended mitigation measures to address these risks, including: (i) recruitment of experienced and qualified staff for the PMO, (ii) provision of financial management support and training under the PIC, (iii) providing training in ADB procedures related to audit, disbursement, and project management. These measures have been incorporated into project design and terms of reference of PIC. Project audit reports will be prepared annually to international accounting standards.

The proposed project fund flow diagram is presented in Figure 8 below. For implementation of the abovementioned activities, GCF will transfer USD 5.0 million to the GCF Account at ADB, in its capacity as Accredited Entity. As

trustee of the GCF Account, ADB will make the GCF Proceeds, in the form of a grant, available to the Executing Entity in accordance with terms and conditions consistent with the Funded Activity Agreement. ADB will conclude a financing agreement with the Republic of Tajikistan, represented by the Ministry of Finance, on the management and use of the proceeds. All disbursements under the loan will be carried out in accordance with ADB's Loan Disbursement Handbook (2017, as amended from time to time). The Hydromet PMO will receive and verify claims for payments from contractors, suppliers and service providers, and consolidate and approve the requests for payment and will forward request for disbursement to ADB. ADB will pay the key suppliers, service providers and contractors via direct payment. An imprest account will be established in a local commercial bank acceptable to ADB for smaller items of expenditure of the PMO and the payment of PMO staff salaries and office operating costs.

Figure 8: Project fund flow



ADB = Asian Development Bank; GCF = Green Climate Fund; PMO = Project Management Office
Source: Asian Development Bank.

A procurement capacity assessment concluded that Hydromet represents a substantial risk. Key sources of risk identified include (i) lack of qualified procurement staff, (ii) lack of experience and unfamiliarity with competitive procurement processes consistent with ADB Procurement Guidelines, (iii) insufficiently centralized record keeping, and (iv) inadequate integrity and anticorruption measures in place. The procurement capacity assessment recommended mitigation measures to address these risks including: (i) recruitment of experienced and qualified staff for the PMO, (ii) provision of procurement support and training under the PIC, (iii) providing training in ADB procedures related to procurement including use of ADB standard bidding documents, and (iv) use of prior review procedures. These measures have been incorporated into project design and terms of reference of PIC.

Procurement methods and thresholds

Procurement is not expected to include highly specialized, complex, or customized civil works or goods. There is a robust local construction industry, with firms likely to be interested in bidding on the construction works for modernizing the Hydromet campus. Monitoring and IT equipment may be procured by shopping or competitive bidding. Detailed design will be conducted during implementation.

The following indicative methods and process thresholds shall apply to procurement of goods and works, and consulting services (see Table 8) below. Bidding documents prepared and procurement initiated by PMO, based on ADB standard bidding documents. Prior review of the bidding documents will be undertaken by ADB. Bid evaluation and bid evaluation reports will be prepared by the PMO with support of the PIC. Bid evaluation will be reviewed by ADB before approval by the government tender board.

Table 7: Procurement methods and thresholds

Method	Threshold
International Competitive Bidding (ICB) for Works	Over \$3,000,000
International Competitive Bidding for Goods	Over \$2,000,000
National Competitive Bidding (NCB) for Works	\$100,000 through \$3,000,000
National Competitive Bidding for Goods	\$100,000 through \$2,000,000
Shopping for Works	Below \$100,000
Shopping for Goods	Below \$100,000

Source: Asian Development Bank

Procurement will follow among others the ADB *Procurement Guidelines* (2015, as amended from time to time) and the *Guidelines on Use of Consultants* (2013, as amended from time to time).

G.1. Risk Assessment Summary

Please provide a summary of main risk factors. Detailed description of risk factors and mitigation measures can be

The following key risks have been identified:

- Political support for project
- Major extreme weather or seismic event
- Staff turnover
- Government financing of operation and maintenance

G.2. Risk Factors and Mitigation Measures

Please describe financial, technical and operational, social and environmental and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures.

Selected Risk Factor 1: Political support for project

Description	Risk category	Level of impact	Probability of risk occurring
Change of government leadership, organization, or priorities at a national and/or local level may result in project refocus, delays, part/whole project cancellation. Legal transformation for Hydromet to more flexibly set salaries to seek and retain additional entrepreneurial income may be hampered. Key Hydromet assets (including modernized campus buildings) may be transferred to other agencies for other use.	Other	High (>20% of project value)	Low

Mitigation Measure(s)

Political risks are inherent in all projects in Central Asia, including Tajikistan. However, climate change adaptation, including improvement of Hydromet services, remain a priority at the highest level of government. ADB has consulted extensively with a broad range of stakeholders on the project, including at the Deputy Prime Minister level, and has received request and support for the project (see Attachment 11). Project covenants will cover legal transformation and Hydromet's exclusive use of modernized assets.

Selected Risk Factor 2: Major extreme weather or seismic event

Description	Risk category	Level of impact	Probability of risk occurring
A major extreme weather or seismic event affect structures or livelihoods in the project areas in Dushanbe and PRB, thereby delaying progress on civil works, capacity building, or stakeholder consultation components, or damaging installed infrastructure or equipment.	Other	Medium (5.1-20% of project value)	Low

Mitigation Measure(s)

A structural integrity assessment has been undertaken of the new Hydromet campus building, confirming adherence to national standards including seismic risk. The campus upgrading will be designed to high standard. The new Hydromet campus is located in the suburbs of Dushanbe, and at insignificant risk of flooding or landslide. ADB will work closely with the relevant government agencies to monitor extreme weather events, particularly annual spring flooding.

Selected Risk Factor 3: Staff turnover

Description	Risk category	Level of impact	Probability of risk occurring
High staff turnover and a limited local human resource base could compromise the PIU and delay implementation.	Technical and operational	Low (<5% of project value)	Medium
Mitigation Measure(s)			
Competitive salaries for PIU staff will be offered. The high priority of the project among stakeholders should make the positions attractive.			

Selected Risk Factor 4: Government financing of operation and maintenance

Description	Risk category	Level of impact	Probability of risk occurring
Hydromet, as a public service entity, will require continued core government funding during and after the project. Insufficient continued government funding to cover operation and maintenance of current and new assets will risk Hydromet's future sustainability and production of high quality outputs.	Financial	Medium (5.1-20% of project value)	Medium
Mitigation Measure(s)			
Hydromet services remain a priority for government, although budget provision is constrained. The World Bank have reported in August 2017 that an increase of budget has been allocated to Hydromet to ensure continued services.			

Selected Risk Factor 5: Financial management and procurement capacity of EE

Description	Risk category	Level of impact	Probability of risk occurring
The FMA for Hydromet as the implementing entity of the subject proposed project concluded that it represents a substantial risk (defined as high impact but low probability risks) due to less effective internal controls, accounting and financial reporting, and funds flow. A procurement capacity assessment concluded that Hydromet represents a substantial risk. Key sources of risk identified include (i) lack of qualified procurement staff, (ii) lack of experience and unfamiliarity with competitive procurement processes consistent with ADB Procurement Guidelines, (iii) insufficiently centralized record keeping, and (iv)	Financial	High (>20% of project value)	Low

inadequate integrity and anticorruption measures in place.			
Mitigation Measure(s)			
The project will undertake the following measures for financial management: (i) recruitment of experienced and qualified staff for the PMO, (ii) provision of financial management support and training under the PIC, (iii) providing training in ADB procedures related to audit, disbursement, and project management. For procurement risk the following measures will be taken: (i) recruitment of experienced and qualified staff for the PMO, (ii) provision of procurement support and training under the PIC, (iii) providing training in ADB procedures related to procurement including use of ADB standard bidding documents, and (iv) use of prior review procedures.			
Other Potential Risks in the Horizon			
Alignment with other donor interventions including World Bank It is anticipated that Hydromet will continue to work with other donors, notably the World Bank, during and after the project. It is understood that the World Bank is considering extensions and follow-on projects for CAHMP. Without appropriate coordination, the respective interventions may not be effective or consistent. ADB is working and will continue to work with Hydromet and other donors to ensure appropriate alignment and synergy with respective future investments and activities. However, implementation of the subject project is unlikely to be affected by other projects, due to the establishment of own project management office.			

H.1. Logic Framework.

H.1.1. Paradigm Shift Objectives and Impacts at the Fund level

Paradigm shift objectives						
Increased climate-resilient sustainable development	This project will support directly PRB area and indirectly the entire country of Tajikistan with improved disaster forecasting and warning, new weather products, and a strengthened Hydromet agency.					
Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Fund-level impacts						
A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions	1.1 Change in expected losses of lives and economic assets (US\$) due to the impact of extreme climate-related disasters in the geographic area of the GCF intervention It is proposed that annual number of households in the selected pilot areas (to be determined during implementation) affected by floods is used as a proxy.	National and district-level reports	To be determined		To be determined	Major extreme weather or seismic events do not occur.
A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions	1.2 Number of males and females benefiting from the adoption of diversified, climate-resilient livelihood options (including fisheries, agriculture, tourism, etc.) It is proposed that the number of households in the PRB subscribing to or otherwise using Hydromet weather services as inputs to	Project reporting and public surveys	To be determined		To be determined	

	their livelihood decision-making.					
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H.1.2. Outcomes, Outputs, Activities and Inputs at Project/Programme level

Expected Result	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term (if applicable)	Final	
Project/programme outcomes	Outcomes that contribute to Fund-level impacts					
A6.0 Increased generation and use of climate information in decision-making	6.2 Use of climate information products/services in decision-making in climate-sensitive sectors Proposed proxy indicator is number of surveyed government and civil society stakeholders 'satisfied' with Hydromet forecasting services	Annual survey	20% (2017 Baseline)	20%	30%	Hydromet development has continued national and local political support.
A6.0 Increased generation and use of climate information in decision-making	6.2 Use of climate information products/services in decision-making in climate-sensitive sectors Proposed proxy indicator is share of Hydromet's annual budget provided by income from fee-based services.	Annual project reports and Hydromet annual financial reporting.	0% (2017 Baseline)	0%	1.5%	Hydromet legal transformation and development has continued national and local political support.
Project/programme outputs	Outputs that contribute to outcomes					

Component A	Hydromet headquarters offices relocated to modernized campus	Quarterly Hydromet project reports	Not applicable (2017 baseline)	Hydromet relocated	Hydromet relocated	
Component B	Hydromet Board of Directors established	Quarterly Hydromet project reports	Not applicable (2017 baseline)	Board of Directors established	Board of Directors established	
Component C	Local flood forecasts issued for pilot PRB districts on Hydromet website at least 36 hours in advance with >40% probability of detection (POD) and <25% false alarm rate (FAR)	Quarterly Hydromet project reports	Not applicable (2017 baseline)	25% POD and 40% FAR	40% POD and 25% FAR	
	At least 500 individuals (of which 30% are female) are trained on flood warnings and preparedness in the PRB	Quarterly Hydromet project reports	0 (2017 baseline)	300 individuals (of which 30% are women)	500 individuals (of which 30% are women)	
Component D	At least one new weather alert product marketed to customers in the PRB by Hydromet (2017 baseline: 0)	Quarterly Hydromet project reports	0 (2017 baseline)	1	1	

	At least 500 individuals (of which 30% are female) are trained on the receipt and use of weather forecasting products in the PRB	Quarterly Hydromet project reports	0 (2017 baseline)	300 individuals (of which 30% are female)	500 individuals (of which 30% are female)	
	IT platform for sale of information and forecasting products established	Quarterly Hydromet project reports	Not applicable (2017 baseline)	Platform established	Platform established	
Activities	Description		Inputs		Description	
Component A: Hydromet campus modernization						
1.1 Design and supervision services engaged	The project will support the modernization of the new Hydromet headquarters campus including main office building, ancillary buildings, and two mixed-use buildings.	1.1.1 Detailed design undertaken for system integration				
		1.1.2 Bidding documents prepared and tendered				
		1.1.3 Construction supervision undertaken				
1.2 Core operational systems modernized		1.2.1 Contract awarded for civil works		See Section C.3.		
		1.2.2 Contract awarded for office furniture				
		1.2.3 Contract awarded for core IT systems				
1.3 Mixed-use buildings constructed		1.3.1 Contract awarded for civil works				
Component B: Legal and organizational transformation						
1.4 Legal reforms implemented	Hydromet’s legal status will be amended to allow for more flexible and autonomous operation including internal reorganization, setting of staff salaries, and seeking	1.4.1 Business development and transition team established		See Section C.3.		
		1.4.2 Project Steering Committee established				

	and retaining additional entrepreneurial income.	1.4.3 Draft government resolution for new legal structure and charter drafted and submitted	
1.5 Institutional framework and quality management system enhanced	The project will enhance Hydromet’s frameworks and systems for strategic planning, management, administration, and control towards ensuring output and development of quality climate services.	1.5.1 Gap analysis and detailed program prepared	See Section C.3.
		1.5.2 Organizational business model and strategic planning updated	
		1.5.3 Management, administration, and control manuals updated or established	
		1.5.4 User feedback and service quality M&E system established	
		1.5.5 Annual stakeholder survey on Hydromet services undertaken	
		1.5.6 Time-bound action plan for ISO 9001 certification developed	
1.6 Capacity building and knowledge-sharing program implemented	Training program for staff and development of regional partnerships.	1.6.1 Training for senior and mid-level Hydromet staff undertaken	See Section C.3.
		1.6.2 Regional and international knowledge partnerships established	
		1.6.3 Regional knowledge sharing event held	
		1.6.4 Annual stakeholder workshop held	
Component C: Forecasting and warning of extreme weather			
1.7 Community awareness and preparedness enhanced	Gender-sensitive consultation with the newly-formed JPRBC and RBOs and other national and local	1.7.1 Gender-sensitive consultation on flood forecasting and warning systems undertaken	See Section C.3.

	government and civil society stakeholders.	1.7.2 Gender-sensitive stakeholder training and awareness campaign on use of flood forecasts and disaster preparedness undertaken	
1.8 Flood forecasting system developed and capacity building undertaken	Hydromet capacity riverine flood forecasting and local flood alerts to support improvements in the water resources management in the PRB area.	1.8.1 Pilot districts selected and detailed design including IT system integration prepared	See Section C.3.
		1.8.2 Flood risk mapping of target areas in PRB prepared	
		1.8.3 Contract awarded for flood forecasting and monitoring equipment	
		1.8.4 Capacity building for Hydromet staff on flood forecasting undertaken	
Component D: Marketing of fee-based services			
1.9 Community awareness and weather product accessibility enhanced	Gender-sensitive consultations and training with farmers and stakeholder organizations including RBOs and WUAs, including awareness campaign.	1.9.1 Gender-sensitive consultation undertaken with farmers and stakeholder organizations in PRB districts on development of weather services	See Section C.3.
		1.9.2 Gender-sensitive stakeholder training and awareness campaign on use and receipt of weather services undertaken	
1.10 Subscription platform and new weather products developed	Platform will be developed for managing the subscription and dissemination of weather information products.	1.10.1 Detailed design undertaken for system integration	See Section C.3.
		1.10.1 Contract awarded for IT systems	
		1.10.2 Website updated and information subscription and	

		delivery system established	
		1.10.3 Capacity building for Hydromet staff on marketing and product development undertaken	

H.2. Arrangements for Monitoring, Reporting and Evaluation

Besides the arrangements (e.g. semi-annual performance reports) laid out in AMA, please provide project/programme specific institutional setting and implementation arrangements for monitoring and reporting and evaluation. Please indicate how the interim/mid-term and final evaluations will be organized, including the timing. Please provide methodologies for monitoring and reporting of the key outcomes of the project/programme.

Monitoring, evaluation, and reporting will follow ADB's Evaluation Policy and the Accreditation Master Agreement concluded between ADB and GCF. The Hydromet PMO established for the project will be responsible for day-to-day project management, monitoring, and reporting. The PMO will prepare quarterly reports on progress, on the current level for all indicators, on the implementation progress and challenges, updated procurement and implementation plan for next 12 months, and the financial status. The PIU will appoint one staff member as focal point for monitoring and evaluation (M&E), and who be supported by M&E specialist under the PIC.

The PMO Director will inform Hydromet and the ADB on an as-needed basis of any delays or difficulties during implementation, so that the appropriate support and corrective measures can be adopted. The Project Director will also ensure that all project staff maintain a high level of transparency, responsibility and accountability in monitoring and reporting project results. The PMO will organize one independent evaluation at the financial close of the program.

In addition:

- A detailed project administration manual will be prepared at the project outset setting out the reporting, monitoring and evaluation activities, responsibilities, and budget.
- ADB will undertake at least two project review missions per year to assess progress of project implementation activities, compliance with covenants and project agreements, and to monitor progress in achieving project outputs and agree on any required modifications.
- ADB will undertake a mid-term review mission through project implementation at a time that ADB and the Government deem it appropriate. The midterm review mission will (i) review institutional, administrative, organizational, technical, environmental, social, economic, procurement, and financial aspects of the project based on the assumptions and risks included in the design and monitoring framework; (ii) review compliance to project covenants; (iii) assess the need to restructure or reformulate the project; and (iv) update the project's design and monitoring framework if restructuring or reformulation is necessary.
- Project implementation will be monitored on an ongoing basis by ADB staff in Manila and Dushanbe.
- Following physical completion of the project, Hydromet will submit a project completion report to ADB.

I. Supporting Documents for Funding Proposal

- ☒ Feasibility Study
- ☒ Appraisal Report or Due Diligence Report with recommendations
- ☒ Map indicating the location of the project/programme (in IEE)
- ☒ Government concurrence letter
- ☒ NDA No-objection Letter
- ☒ Integrated Economic and Financial Model that provides sensitivity analysis of critical elements

Attached Documents:

1. Institutional Transformation
2. Legal Structure
3. Campus Modernization
4. Financial Management Assessment
5. Summary Poverty Reduction and Social Strategy
6. Gender Action Plan
7. Project Procurement Risk Assessment
8. Financial Assessment
9. Economic Assessment
10. Supplementary Initial Environmental Examination (IEE)
11. Government Concurrence
12. NDA No Objection Letter
13. Poverty, Social, and Gender Analysis
14. Water Resources Management in Pyanj River Basin project summary

- ☒ Timetable of project/programme implementation
- ☒ Appraisal document (Report and Recommendations of the President) of the WRMPRB project
- ☒ ADB Tajikistan Country Operating Business Plan (2018-2020)



No-objection letter issued by the national designated authority

**КУМИТАИ
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**КОМИТЕТ
ОХРАНЫ ОКРУЖАЮЩЕЙ СРЕДЫ
ПРИ ПРАВИТЕЛЬСТВЕ
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**COMMITTEE OF ENVIRONMENTAL PROTECTION
UNDER THE GOVERNMENT OF THE REPUBLIC OF TAJIKISTAN**

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№ 1117-03/1877 аз « 12 » 10 соли 2016
Ба № _____ аз « _____ » _____ соли 2016

To: The Green Climate Fund (GCF)

Dushanbe, 12.10.2017

Re: Funding proposal for the GCF by the Asian Development Bank (ADB) regarding the Water Resources Management in the Pyanj River Basin (Additional Financing) - Institutional Development of State Agency for Hydrometeorology of Tajikistan

Dear Madam, Sir,

We refer to the Water Resources Management in the Pyanj River Basin (Additional Financing) Institutional Development of State Agency for Hydrometeorology of Tajikistan as included in the funding proposal submitted by the Asian Development Bank (ADB) to us on 20 September 2017.

The undersigned is the duly authorized representative of the Committee on Environmental Protection under the Government of the Republic of Tajikistan, the National Designated Authority/focal point of Tajikistan.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the project as included in the funding proposal.

By communicating our no-objection, it is implied that:

(a) The government of Tajikistan has no-objection to the project as included in the funding proposal;

(b) The project as included in the funding proposal is in conformity with Tajikistan's national priorities, strategies and plans;

(c) In accordance with the GCF's environmental and social safeguards, the project as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the project as included in the funding proposal has been duly followed. We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Name: Mr. Khayrullo Ibodzoda

Title: Chairman of the Committee of Environmental Protection

Environmental and social report(s) disclosure

Basic project/programme information	
Project/programme title	Institutional Development of the State Agency of Hydrometeorology of Tajikistan
Accredited entity	Asian Development Bank
Environmental and social safeguards (ESS) category	Category B

Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	2018-01-25
Language(s) of disclosure	English, Russian
Link to disclosure	https://www.adb.org/projects/documents/taj-47181-003-iee A simplified ESIA consistent with the requirements for a category B project is contained in the Initial Environmental Examination (IEE)
Other link(s)	https://www.adb.org/ru/projects/documents/taj-47181-003-iee
Environmental and Social Impact Assessment (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	2018-01-25
Language(s) of disclosure	English, Russian
Link to disclosure	https://www.adb.org/projects/documents/taj-47181-003-iee An ESMP consistent with the requirements for a category B project is contained in the Initial Environmental Examination (IEE)
Other link(s)	https://www.adb.org/ru/projects/documents/taj-47181-003-iee
Resettlement Action Plan (RAP) (if applicable)	
Date of disclosure on accredited entity's website	Not applicable
Any other relevant ESS reports and/or disclosures (if applicable)	
Description of report/disclosure	Not applicable